AI4I 2020 Predictive Maintenance Knowledge Manual

Chapter 1: Dataset Overview and Machine Specifications

Project: Faithfulness-Verified Multi-Source Agentic RAG for Industrial Predictive Maintenance

Dataset: AI4I 2020 Predictive Maintenance Dataset

Dataset Source: UCI Machine Learning Repository

1.1 Introduction

Predictive maintenance is an approach used in industrial environments to identify potential equipment failures before they cause unexpected machine downtime. It uses operational measurements, historical data and analytical models to monitor equipment conditions and support maintenance decisions.

The AI4I 2020 Predictive Maintenance Dataset is a synthetic industrial dataset designed to represent predictive maintenance scenarios. It simulates operating conditions associated with an industrial milling machine and provides measurements related to temperature, rotational speed, torque and tool wear.

The dataset contains operational measurements and six target variables: one overall machine failure indicator and five individual failure-mode indicators.

This manual uses the dataset as the foundation for explaining machine operating parameters, failure mechanisms, diagnostic information and proposed maintenance procedures.

Important: AI4I 2020 is a synthetic research dataset. It is not a manufacturer's operating or maintenance manual, and it does not specify a particular commercial machine model.

1.2 Dataset Characteristics

Property	Description
Dataset name	AI4I 2020 Predictive Maintenance Dataset
Year	2020
Source	UCI Machine Learning Repository
Application domain	Industrial predictive maintenance
Simulated equipment	Industrial milling machine

Total records	10,000
Total columns	14
Data type	Multivariate
Data characteristics	Synthetic operational measurements
Primary task	Machine failure prediction
Additional task	Individual failure-mode prediction
Missing values	None in the original UCI dataset
License	Creative Commons Attribution 4.0 (CC BY 4.0)

The dataset is designed to support research involving classification, regression and predictive maintenance.

1.3 Simulated Machine Operating Environment

The AI4I dataset represents a simulated industrial milling process. Its records describe operating conditions using five primary numerical measurements and one categorical product-quality indicator.

The simulated operating environment includes:

- Ambient air temperature.
- Process temperature.
- Rotational speed.
- Applied torque.
- Cutting tool wear.
- Product quality category.

These parameters represent different aspects of the machine's thermal and mechanical operating conditions.

The dataset does not provide a complete physical machine specification, manufacturer details, component dimensions, rated motor capacity or actual sensor calibration information. Consequently, its measurements should be interpreted as simulated operational variables rather than verified specifications of a particular industrial machine.

1.4 Product Type Classification

The dataset contains three product-quality variants:

Product type	Description	Approximate distribution
L	Low-quality variant	50%
M	Medium-quality variant	30%
H	High-quality variant	20%

The product type is relevant because the simulated failure-generation mechanisms use different thresholds for certain failure modes.

For example, the overstrain failure threshold depends on the product variant.

Product type is therefore an important categorical input for machine learning and failure diagnosis.

1.5 Sensor Parameters and Operating Variables

The following sections describe the five numerical operating measurements included in the dataset.

1.5.1 Air Temperature [K]

Air temperature represents the surrounding environmental temperature associated with the simulated machine operation.

- Column name: Air temperature [K]
- Data type: Continuous numerical
- Measurement unit: Kelvin (K)

Engineering significance:

Ambient temperature influences the thermal environment in which industrial machinery operates. Changes in surrounding temperature can affect heat transfer between machine components and their environment.

In the AI4I dataset, air temperature is generated using a random-walk process and normalized around 300 K with a standard deviation of approximately 2 K.

Relevance to predictive maintenance:

Air temperature is considered together with process temperature when evaluating heat dissipation conditions.

1.5.2 Process Temperature [K]

Process temperature represents the operating temperature associated with the simulated industrial process.

- Column name: Process temperature [K]
- Data type: Continuous numerical
- Measurement unit: Kelvin (K)

Engineering significance:

Process temperature provides information about the thermal condition of the operating process. Elevated or abnormal process temperatures may indicate changes in operating conditions or heat-management performance.

In the dataset, process temperature is generated relative to air temperature, with an approximate nominal difference of 10 K and additional variation.

Relevance to predictive maintenance:

Process temperature is used together with air temperature to calculate the temperature difference, which is relevant to the Heat Dissipation Failure (HDF) mechanism.

1.5.3 Rotational Speed [rpm]

Rotational speed describes the number of revolutions performed by the machine's rotating component per minute.

- Column name: Rotational speed [rpm]
- Data type: Integer numerical
- Measurement unit: Revolutions per minute (rpm)

Engineering significance:

Rotational speed is an important operating parameter in rotating machinery. Changes in speed can affect mechanical loading, power requirements, heat generation and process performance.

The AI4I dataset generates rotational speed using a power-related model with normally distributed variation.

Relevance to predictive maintenance:

Rotational speed contributes to the evaluation of:

- Heat Dissipation Failure (HDF).
- Power Failure (PWF).
- Overall Machine Failure.

1.5.4 Torque [Nm]

Torque represents the rotational force applied to the machine's rotating system.

- Column name: Torque [Nm]
- Data type: Continuous numerical
- Measurement unit: Newton-metre (Nm)

Engineering significance:

Torque represents the mechanical loading experienced by the rotating system. Higher torque indicates greater rotational force requirements.

The dataset generates torque values around a nominal mean of approximately 40 Nm, with a standard deviation of approximately 10 Nm.

Relevance to predictive maintenance:

Torque is important for evaluating:

- Power Failure (PWF).
- Overstrain Failure (OSF).
- Mechanical operating conditions.

Torque is also useful when calculating estimated mechanical power using rotational speed.

1.5.5 Tool Wear [min]

Tool wear represents the accumulated usage time of the cutting tool during the simulated manufacturing process.

- Column name: Tool wear [min]
- Data type: Integer numerical
- Measurement unit: Minutes

Engineering significance:

Cutting tools gradually experience wear during machining operations. Excessive wear can affect machining quality, cutting resistance and operational reliability.

In the AI4I dataset, tool wear increases according to the product type:

- L variant: 2 minutes of additional tool wear.
- M variant: 3 minutes of additional tool wear.

- H variant: 5 minutes of additional tool wear.

Relevance to predictive maintenance:

Tool wear is particularly important for:

- Tool Wear Failure (TWF).
- Overstrain Failure (OSF).
- Overall Machine Failure.

It is one of the primary parameters used to identify tool-related degradation conditions.

1.6 Dataset Column Classification

The original dataset contains 14 columns divided into input variables, identifiers and target variables.

Column	Category	Description
UDI	Identifier	Unique record identifier
Product ID	Identifier	Product-specific identifier
Type	Input feature	Product quality variant
Air temperature [K]	Input feature	Ambient temperature
Process temperature [K]	Input feature	Process operating temperature
Rotational speed [rpm]	Input feature	Rotational speed
Torque [Nm]	Input feature	Applied rotational force
Tool wear [min]	Input feature	Accumulated tool usage
Machine failure	Target	Overall machine failure indicator
TWF	Target	Tool Wear Failure
HDF	Target	Heat Dissipation Failure
PWF	Target	Power Failure
OSF	Target	Overstrain Failure
RNF	Target	Random Failure

Identifier handling: UDI and Product ID are identifiers rather than physical operating measurements. They should generally be excluded from the machine learning input features to avoid learning irrelevant record-specific patterns.

1.7 Failure Classification Overview

The AI4I dataset defines five individual failure mechanisms.

Failure code	Full name	Primary parameters
TWF	Tool Wear Failure	Tool wear
HDF	Heat Dissipation Failure	Air temperature, process temperature, rotational speed
PWF	Power Failure	Torque, rotational speed
OSF	Overstrain Failure	Torque, tool wear, product type
RNF	Random Failure	Random event

Each failure mode is represented by a binary target:

- 0: The corresponding failure mode is not labelled for that record.
- 1: The corresponding failure mode is labelled for that record.

Detailed failure-generation conditions, technical explanations, inspection procedures and maintenance recommendations are provided in their respective chapters.

1.8 Overall Machine Failure Indicator

The Machine failure column represents the overall failure status of the simulated machine.

It is a binary target:

- 0: No machine failure is labelled.
- 1: At least one of the five failure modes is labelled.

Conceptually:

Machine failure = TWF OR HDF OR PWF OR OSF OR RNF

This means that overall machine failure is an aggregate target derived from the individual failure-mode indicators.

A record may have more than one failure-mode indicator set to 1. Therefore, the five failure modes should not be treated as mutually exclusive classes.

The overall machine failure target is useful for binary classification, while the individual failure-mode targets support more detailed fault diagnosis.

1.9 Dataset Limitations

The following limitations should be considered when developing predictive maintenance models using AI4I 2020.

1. Synthetic data generation

The dataset contains simulated measurements and generated failure labels. Its behaviour may not fully represent real industrial operating conditions.

2. Limited equipment specifications

The dataset does not identify a specific manufacturer, machine model or complete mechanical configuration.

3. Limited sensor information

The dataset contains five primary numerical operating measurements. It does not provide detailed vibration signals, acoustic measurements, oil analysis or raw sensor waveforms.

4. Imbalanced failure classes

Failure-mode indicators have significantly fewer positive records than normal records. Models should therefore be evaluated using precision, recall, F1-score and precision-recall AUC, rather than accuracy alone.

5. Random failure limitation

Random Failure (RNF) is generated independently of the available operating measurements. Consequently, it cannot be reliably predicted from the other sensor variables using ordinary supervised learning.

6. Limited real-world generalization

A model achieving high performance on AI4I does not automatically guarantee similar performance on actual industrial equipment. Validation using real operational data is necessary before practical deployment.

1.10 Role of This Dataset in the Proposed Research

In the proposed Faithfulness-Verified Multi-Source Agentic RAG architecture, AI4I 2020 provides the structured operational data used for machine failure prediction.

Its role can be summarized as follows:

1. **Structured data processing:** Clean and preprocess the sensor measurements.

2. **Machine learning:** Predict overall machine failure and individual failure modes.
3. **Explainable AI:** Use SHAP to identify features influencing individual predictions.
4. **Knowledge retrieval:** Retrieve relevant failure definitions, technical explanations and maintenance procedures from the project knowledge base.
5. **Agentic diagnosis:** Combine model predictions, SHAP explanations and retrieved information to generate a diagnosis.
6. **Faithfulness verification:** Verify that generated technical claims are supported by the retrieved evidence.

The AI4I dataset provides the operational evidence for the prediction stage, while the maintenance knowledge base supplies external technical information for diagnosis and recommendations.

1.11 References

1. UCI Machine Learning Repository. (2020). *AI4I 2020 Predictive Maintenance Dataset*. DOI: 10.24432/C5HS5C. https://doi.org/10.24432/C5HS5C
2. Matzka, S. (2020). *Explainable Artificial Intelligence for Predictive Maintenance Applications*. International Conference on Artificial Intelligence for Industries (AI4I).
3. UCI Machine Learning Repository. *AI4I 2020 Dataset – Additional Variable Information and Failure Generation Rules*. https://archive.ics.uci.edu/dataset/601/ai4i+2020

Chapter 2: Tool Wear Failure (TWF) – Technical Analysis and Maintenance Procedures

Manual: AI4I 2020 Predictive Maintenance Knowledge Manual
Failure Category: Tool Wear Failure (TWF)
Dataset: AI4I 2020 Predictive Maintenance Dataset
Application: Simulated industrial milling machine

2.1 Introduction

Tool Wear Failure (TWF) is a failure mechanism associated with the gradual deterioration of a cutting tool during machining operations.

In industrial milling processes, cutting tools are subjected to repeated mechanical interactions with workpiece materials. Over time, these interactions can cause wear on the cutting edges and other tool surfaces.

Excessive tool wear can affect machining accuracy, surface finish, cutting performance and operational reliability. If a worn tool continues to operate without appropriate inspection or replacement, it may contribute to poor manufacturing quality or unexpected process interruption.

The AI4I 2020 Predictive Maintenance Dataset incorporates Tool Wear Failure as one of its five simulated failure mechanisms.

This chapter describes the dataset-defined TWF mechanism and provides additional engineering knowledge that can support fault diagnosis and maintenance recommendations.

2.2 Definition of Tool Wear Failure

Tool Wear Failure occurs when a cutting tool reaches a specified wear condition and subsequently experiences a simulated failure event.

Tool wear is represented in the AI4I dataset by the variable:

Tool wear [min]

This variable measures the accumulated tool usage time in minutes.

As the simulated manufacturing process progresses, tool wear increases according to the product type.

The dataset uses a randomly selected tool-wear threshold between 200 and 240 minutes for its TWF failure-generation mechanism.

At the selected wear threshold, the simulation can either replace the tool or assign a tool wear failure event.

Important distinction: The TWF label should not be interpreted as a simple deterministic rule stating that every record with tool wear above 200 minutes represents a failure. The dataset uses a simulated replacement/failure mechanism.

2.3 Dataset-Defined TWF Generation Mechanism

According to the UCI AI4I 2020 dataset documentation, the TWF mechanism is generated using a randomly selected tool-wear threshold between 200 and 240 minutes.

The simulation includes tool replacement and failure outcomes at the selected threshold.

The dataset documentation describes 120 such events, of which 69 result in tool replacement and 51 result in failure in its documented generation process.

The exact label distribution should be verified against the particular CSV version being used in an experiment.

2.3.1 Tool Wear Accumulation

The dataset models tool wear accumulation according to product quality type.

Product type	Tool wear increment
L	2 minutes
M	3 minutes
H	5 minutes

This means that different product variants contribute different amounts of simulated tool wear during processing.

For example:

- An L-type product adds 2 minutes of wear.
- An M-type product adds 3 minutes of wear.
- An H-type product adds 5 minutes of wear.

2.3.2 TWF Failure Condition

The dataset uses a randomly selected wear threshold:

TWF threshold $\in [200, 240]$ minutes

When the simulated tool reaches this threshold, the tool may be replaced or experience a failure event.

This is a dataset-generation mechanism rather than a universal industrial tool-replacement limit.

2.3.3 Interpretation for Predictive Maintenance

A machine learning model trained on AI4I data learns statistical relationships between tool wear, other operating parameters and the TWF label.

The model estimates whether a record is associated with a tool wear failure condition.

It does not directly observe the physical condition of a cutting edge or measure actual tool degradation.

Therefore, model predictions should be interpreted as estimates based on the available simulated sensor variables.

2.4 Relevant Sensor Parameters

The following variables are particularly relevant when analyzing TWF.

2.4.1 Tool Wear [min]

Dataset column: Tool wear [min]

Unit: Minutes

Tool wear represents accumulated tool usage.

Engineering significance:

- Higher accumulated usage can indicate increased exposure to wear.
- Tool wear can influence machining performance.
- Excessive wear may contribute to cutting-edge degradation.
- Tool condition inspection can help identify when replacement is necessary.

Role in TWF:

Tool wear is the primary variable associated with the dataset's TWF generation mechanism.

2.4.2 Torque [Nm]

Dataset column: Torque [Nm]

Unit: Newton-metre

Torque represents the rotational force applied to the machine's rotating system.

Engineering significance:

- Torque provides information about mechanical loading.
- Increased cutting resistance may affect the torque required during machining.
- Abnormal torque behaviour can indicate changes in operating conditions.

Role in TWF:

Torque is not the primary threshold variable in the AI4I TWF generation rule. However, it may provide additional information for machine learning models and for engineering analysis.

A high torque measurement alone does not establish that tool wear failure has occurred.

2.4.3 Rotational Speed [rpm]

Dataset column: Rotational speed [rpm]

Unit: Revolutions per minute

Rotational speed describes the operating speed of the rotating system.

Engineering significance:

- Cutting speed influences machining conditions.
- Different tool materials and workpiece materials require suitable operating speeds.
- Inappropriate operating conditions can contribute to accelerated tool degradation.

Role in TWF:

Rotational speed is not a direct condition in the AI4I TWF generation rule. It can still be considered as supporting information during predictive analysis.

2.4.4 Product Type

Dataset column: Type

Possible values:

- L: Low-quality variant.
- M: Medium-quality variant.
- H: High-quality variant.

Role in TWF:

Product type determines the simulated tool-wear increment per processed product.

It is therefore relevant to understanding the rate at which accumulated tool wear changes in the dataset.

2.5 Possible Engineering Symptoms of Tool Wear

The AI4I dataset does not contain direct measurements of physical tool damage or machining surface quality.

However, general industrial machining knowledge identifies several possible indicators of cutting-tool degradation.

Possible symptom	Engineering interpretation
Poor surface finish	A worn cutting edge may produce an inferior machined surface
Increased cutting resistance	Tool degradation may affect cutting performance
Abnormal vibration	Tool condition or cutting instability may contribute to vibration
Increased cutting temperature	Friction and cutting conditions may influence heat generation
Dimensional inaccuracy	Tool degradation can affect machining precision
Visible cutting-edge damage	Physical inspection may reveal wear, chipping or deformation

These symptoms are general engineering indicators. They are not directly represented as separate sensor columns or labelled observations in AI4I.

2.6 Possible Contributing Factors

Tool wear can be influenced by multiple operating and material-related factors.

2.6.1 Mechanical Friction

Repeated contact between the cutting tool and workpiece material can contribute to progressive material loss from the tool surface.

2.6.2 Excessive Mechanical Loading

High mechanical loading can increase stress on the cutting tool and contribute to degradation.

2.6.3 Inappropriate Cutting Conditions

Unsuitable cutting speeds, feed rates or cutting depths may accelerate wear depending on the tool and material combination.

2.6.4 Thermal Effects

High temperatures at the cutting interface can contribute to tool material degradation.

2.6.5 Inadequate Lubrication or Cooling

In machining processes that require coolant or lubrication, inadequate supply can affect cutting temperature, friction and tool life.

Dataset limitation: AI4I does not directly measure all these contributing factors. They are included as general engineering knowledge, not as proven causes of individual AI4I records.

2.7 Tool Inspection Procedures

The following inspection procedures are proposed as general guidance for machining equipment. Actual inspection methods must follow the machine manufacturer's instructions and the applicable safety procedures.

2.7.1 Visual Inspection

Inspect the cutting tool for:

- Cutting-edge wear.
- Surface damage.
- Chipping or cracks.
- Deformation.
- Excessive material loss.

2.7.2 Tool Condition Assessment

Evaluate the tool's physical condition and compare it with the applicable tool manufacturer's acceptance criteria.

2.7.3 Operating Parameter Review

Review available operational measurements:

- Accumulated tool wear.
- Torque.
- Rotational speed.
- Process temperature.

Compare the readings with historical operating behaviour and equipment-specific operating limits.

2.7.4 Tool Replacement Assessment

If inspection identifies unacceptable wear or damage, determine whether the tool requires replacement.

Replacement decisions should be based on physical inspection and manufacturer specifications rather than relying exclusively on the AI4I 200–240-minute simulated threshold.

2.8 Troubleshooting Procedures

The following table provides proposed troubleshooting guidance for suspected tool wear problems.

Observation	Possible explanation	Suggested action
High accumulated tool wear	Extended tool usage	Inspect tool condition
Poor machining surface	Possible cutting-edge degradation	Inspect cutting edges and machining conditions
Increased cutting resistance	Possible tool wear or process change	Review torque and inspect the tool
Visible tool damage	Physical deterioration	Follow equipment procedures for tool replacement
Repeated tool wear warnings	Possible unsuitable tool-life management	Review inspection and replacement intervals

2.8.1 General Troubleshooting Workflow

Step 1: Identify the warning

Review the predictive maintenance system's TWF prediction and associated confidence or probability.

Step 2: Examine the sensor measurements

Review accumulated tool wear, torque and rotational speed.

Step 3: Inspect the cutting tool

Conduct an appropriate physical inspection following machine safety procedures.

Step 4: Determine the corrective action

Depending on the inspection results, possible actions include tool replacement, operating-condition adjustment or further investigation.

Step 5: Record the maintenance activity

Document the observed condition, inspection findings, corrective action and tool replacement details.

Step 6: Resume operation safely

Resume operation only after the equipment and tooling have been assessed as safe for use.

2.9 Preventive Maintenance Recommendations

Preventive maintenance aims to reduce unexpected interruptions and maintain consistent machining performance.

The following recommendations are proposed for general tool-wear management.

2.9.1 Periodic Tool Inspection

Inspect cutting tools at intervals determined by equipment specifications, tool characteristics and operating conditions.

2.9.2 Tool-Life Monitoring

Monitor accumulated tool usage and maintain records of inspection and replacement events.

2.9.3 Condition-Based Replacement

Consider replacing tools based on actual condition, manufacturer limits and observed machining performance.

2.9.4 Operating Parameter Monitoring

Monitor torque, rotational speed and process temperature to identify unusual operating behaviour.

2.9.5 Maintenance Record Management

Maintain records containing:

- Tool identification.
- Installation date.
- Accumulated usage.
- Inspection results.

- Replacement date.
- Observed tool damage.
- Corrective maintenance actions.

2.9.6 Appropriate Cutting Conditions

Follow the recommended cutting parameters for the tool material, workpiece material and machine configuration.

2.10 Relationship Between TWF and Other Failure Modes

Although the AI4I dataset defines five separate failure indicators, more than one failure mode may be labelled for the same record.

Tool wear is particularly relevant to Overstrain Failure (OSF).

Failure mode	Relationship with tool wear
TWF	Directly associated with the simulated tool-wear threshold
HDF	Not directly determined by tool wear
PWF	Not directly determined by tool wear
OSF	Uses tool wear together with torque and product type
RNF	Generated independently of the operating parameters

For OSF, the AI4I dataset uses a product of tool wear and torque, with a threshold that varies by product type.

Consequently, a machine record may be associated with both TWF and OSF.

The predictive maintenance system should therefore support multiple failure indicators rather than assuming that every machine can experience only one failure mode.

2.11 Role of TWF in the Proposed Machine Learning System

The TWF label is used as an individual binary prediction target.

The machine learning model receives the available input features:

- Type.
- Air temperature.
- Process temperature.
- Rotational speed.

- Torque.
- Tool wear.

The model generates a TWF prediction and an associated probability estimate.

For example, an illustrative output might be:

Output	Example
Predicted failure type	TWF
Predicted probability	82%
Tool wear	215 minutes
Torque	48 Nm
Rotational speed	1450 rpm

These values are illustrative and do not represent an actual prediction from the dataset.

2.12 Role of SHAP in TWF Diagnosis

SHAP (SHapley Additive exPlanations) is used to explain the contribution of individual input features to the trained model's prediction.

For TWF, SHAP can help identify whether tool wear, torque, rotational speed, temperature or product type influenced the model output.

An illustrative explanation could be:

Feature	Example SHAP contribution	Interpretation
Tool wear	+0.25	Pushes the prediction towards TWF
Torque	+0.10	Pushes the prediction towards TWF
Rotational speed	-0.04	Pushes the prediction away from TWF
Process temperature	+0.03	Pushes the prediction towards TWF

These values are illustrative, not measured results.

SHAP contributions explain the model's behaviour. They do not establish the physical cause of a tool failure.

2.13 Role of RAG in TWF Diagnosis

The maintenance knowledge base contains the technical information described in this chapter.

When the ML model predicts TWF, the diagnosis agent can retrieve relevant sections from the manual.

Examples of useful retrieval topics include:

- Tool wear definition.
- Tool-wear accumulation.
- Tool inspection procedures.
- Cutting-edge damage.
- Tool replacement guidance.
- Preventive maintenance recommendations.

The retrieved content can be combined with the ML prediction and SHAP explanation to generate a maintenance report.

Example Internal Retrieval Query

Failure category: Tool Wear Failure (TWF)

Relevant parameters:

- Tool wear.
- Torque.
- Rotational speed.

Retrieval request:

"Retrieve technical information about tool wear, cutting-tool inspection, possible wear symptoms, troubleshooting procedures and preventive maintenance recommendations for an industrial milling machine."

Example Retrieved Context

The knowledge base may retrieve information explaining that accumulated tool wear is associated with cutting-tool usage and that physical inspection should be performed to evaluate cutting-edge condition.

It may also retrieve the recommended general inspection and replacement guidance documented in this chapter.

2.14 Faithfulness Verification Requirements

The generated diagnosis must be supported by the retrieved technical evidence.

The faithfulness verifier should distinguish the following claims:

Claim type	Required supporting evidence
Dataset-defined TWF threshold	AI4I dataset documentation
Tool-wear measurement	AI4I variable documentation
Cutting-edge wear symptoms	Engineering reference
Inspection procedures	Technical maintenance documentation
Tool replacement recommendations	Manufacturer guidance or clearly labelled proposed recommendations
Model prediction	Actual ML model output
Feature contribution	Actual SHAP output

The system must not generate unsupported claims such as declaring that a cutting tool is physically damaged solely because the model predicts TWF.

A predicted failure is an indication requiring further investigation, not proof of physical damage.

2.15 Knowledge Base Retrieval Metadata

The following metadata structure is recommended for storing TWF-related document chunks.


```
{
  "document_id": "AI4I_MANUAL_002",
  "chapter": 2,
  "failure_type": "TWF",
  "failure_name": "Tool Wear Failure",
  "section": "Inspection Procedures",
  "equipment": "Simulated Industrial Milling Machine",
  "source_type": "Proposed Maintenance Guidance",
```

"content": "Inspect the cutting tool for visible wear, "
"chipping, cracks and deformation."
}

Suggested section categories:

Category	Description
Failure_Definition	TWF explanation
Dataset_Rule	AI4I failure-generation mechanism
Sensor_Parameters	Relevant operating measurements
Engineering_Symptoms	Possible physical indicators
Contributing_Factors	General wear-related factors
Inspection	Tool inspection procedures
Troubleshooting	Corrective actions
Preventive_Maintenance	Preventive recommendations
Limitations	Dataset and prediction limitations

These categories can be used as metadata filters during retrieval.

2.16 Chapter Summary

Tool Wear Failure is a simulated failure mechanism in the AI4I 2020 Predictive Maintenance Dataset.

Its generation is associated with a randomly selected tool-wear threshold between 200 and 240 minutes and a simulated tool replacement/failure process.

Tool wear is the primary variable associated with this failure mechanism. Torque, rotational speed and product type can provide additional context for predictive analysis and engineering interpretation.

The proposed maintenance procedures focus on tool inspection, condition assessment, appropriate replacement decisions, operating-condition monitoring and maintenance record management.

Within the proposed research architecture:

1. The ML model estimates the likelihood of TWF.
2. SHAP explains the features influencing the prediction.

3. RAG retrieves relevant tool-wear knowledge and maintenance procedures.
4. The LLM generates a diagnosis and recommended actions.
5. The faithfulness verifier checks the generated claims against the retrieved evidence.

This integration connects structured machine learning predictions with external technical knowledge to support explainable and evidence-grounded predictive maintenance.

2.17 References

1. UCI Machine Learning Repository. (2020). *AI4I 2020 Predictive Maintenance Dataset*. https://doi.org/10.24432/C5HS5C
2. Matzka, S. (2020). *Explainable Artificial Intelligence for Predictive Maintenance Applications*. Third International Conference on Artificial Intelligence for Industries (AI4I).
3. ISO 13381-1:2015. *Condition monitoring and diagnostics of machines — Prognostics — Part 1: General guidelines*. Consult the standard for general prognostics principles.
4. ISO 17359:2018. *Condition monitoring and diagnostics of machines — General guidelines*. Consult the standard for condition-monitoring practices.

Source distinction: The TWF generation description is based on AI4I documentation. The engineering symptoms, inspection procedures and maintenance recommendations are general proposed guidance and should be validated against the actual machine manufacturer's specifications before industrial use.

AI4I 2020 Predictive Maintenance Knowledge Manual

Chapter 3: Heat Dissipation Failure (HDF) – Technical Analysis and Maintenance Procedures

Manual: AI4I 2020 Predictive Maintenance Knowledge Manual
Failure Category: Heat Dissipation Failure (HDF)
Dataset: AI4I 2020 Predictive Maintenance Dataset
Application: Simulated Industrial Milling Machine

3.1 Introduction

Heat Dissipation Failure (HDF) is a failure mechanism associated with insufficient heat dissipation during machine operation.

Industrial machinery generates heat through mechanical movement, friction, electrical losses and machining processes. Effective heat transfer is necessary to maintain suitable operating temperatures and prevent excessive thermal stress.

When heat generation exceeds the machine's ability to dissipate heat, operating temperatures may increase and affect equipment performance.

The AI4I 2020 Predictive Maintenance Dataset includes HDF as one of its five simulated failure mechanisms.

In this dataset, HDF is identified using a combination of temperature difference and rotational speed.

This chapter explains the dataset-defined HDF condition, the relevant sensor measurements, possible engineering symptoms, inspection procedures, troubleshooting methods and preventive maintenance recommendations.

3.2 Definition of Heat Dissipation Failure

Heat Dissipation Failure occurs when the simulated machine satisfies specific thermal and rotational operating conditions associated with insufficient heat dissipation.

The AI4I dataset uses two important operating measurements:

1. Difference between process temperature and air temperature.
2. Rotational speed of the machine.

The dataset defines HDF using the following condition:

- Temperature difference is below 8.6 K.
- Rotational speed is below 1380 rpm.

Both conditions must be satisfied for the dataset-defined HDF rule.

Important: This is a synthetic failure-generation rule. It should not be interpreted as a universal industrial overheating threshold or a general rule for all machines.

3.3 Dataset-Defined HDF Generation Mechanism

According to the AI4I 2020 dataset documentation, HDF is generated when the temperature difference between the process and surrounding air becomes sufficiently small while rotational speed is below the specified threshold.

3.3.1 Temperature Difference Calculation

The temperature difference is calculated as:

Temperature Difference = Process Temperature - Air Temperature

Mathematically:

[
\Delta T = T_{\text{process}} - T_{\text{air}}
]

Where:

- (\Delta T) = Temperature difference in Kelvin.
- (T_{\text{process}}) = Process temperature in Kelvin.
- (T_{\text{air}}) = Air temperature in Kelvin.

For example:

If:

- Air temperature = 303 K
- Process temperature = 311 K

Then:

[
\Delta T = 311 - 303 = 8K
]

The calculated temperature difference is 8 K.

3.3.2 HDF Failure Condition

The AI4I HDF generation rule can be represented as:

[
HDF =
\begin{cases}
1, & \Delta T < 8.6K \text{ and RPM} < 1380\
0, & \text{otherwise}
\end{cases}
]

The two conditions must occur together.

Parameter	Dataset condition
Temperature difference	Below 8.6 K
Rotational speed	Below 1380 rpm
Logical condition	Both conditions must be satisfied

3.3.3 Example of HDF Detection

Consider three hypothetical operating scenarios.

Scenario	Temperature difference	RPM	HDF condition
A	8.0 K	1300	Satisfied
B	9.0 K	1300	Not satisfied
C	8.0 K	1450	Not satisfied

Interpretation:

- Scenario A satisfies both conditions and is associated with HDF under the dataset rule.
- Scenario B does not satisfy the temperature-difference condition.
- Scenario C does not satisfy the rotational-speed condition.

This demonstrates why both sensor measurements must be considered together.

3.4 Relevant Sensor Parameters

The following measurements are important when analyzing HDF.

3.4.1 Air Temperature [K]

Dataset column: Air temperature [K]

Unit: Kelvin (K)

Air temperature represents the surrounding environmental temperature associated with the simulated machine.

Engineering significance:

- It represents the ambient thermal condition.
- It influences the temperature gradient between the machine process and its surroundings.
- It provides a reference for calculating temperature difference.
- Changes in ambient temperature can influence heat-transfer conditions.

Role in HDF:

Air temperature is one of the two measurements used to calculate the temperature difference in the AI4I HDF generation rule.

3.4.2 Process Temperature [K]

Dataset column: Process temperature [K]

Unit: Kelvin (K)

Process temperature represents the operating temperature associated with the simulated manufacturing process.

Engineering significance:

- It provides information about the thermal condition of the machine process.
- It can help identify changes in operating temperature.
- It is useful for monitoring thermal behaviour.
- It can be compared with surrounding air temperature to understand the temperature gradient.

Role in HDF:

Process temperature is directly involved in calculating the temperature difference used by the AI4I HDF rule.

3.4.3 Rotational Speed [rpm]

Dataset column: Rotational speed [rpm]

Unit: Revolutions per minute (rpm)

Rotational speed represents the number of revolutions performed by the machine's rotating system per minute.

Engineering significance:

- It describes the operating speed of rotating components.
- It influences mechanical operating conditions.
- It can affect friction, heat generation and power requirements.
- It provides information about changes in machine operation.

Role in HDF:

Rotational speed is the second condition in the AI4I HDF generation mechanism.

The dataset uses a threshold of 1380 rpm.

3.4.4 Torque [Nm]

Dataset column: Torque [Nm]

Unit: Newton-metre (Nm)

Torque represents the rotational force applied to the machine's rotating system.

Engineering significance:

- It provides information about mechanical loading.
- It can help identify changes in operating conditions.
- It is useful when evaluating the relationship between mechanical load and heat generation.

Role in HDF:

Torque is not directly included in the AI4I HDF generation condition. However, it can provide additional information to a machine learning model.

3.4.5 Tool Wear [min]

Dataset column: Tool wear [min]

Unit: Minutes

Tool wear represents accumulated tool usage time.

Engineering significance:

- It indicates the usage history of the cutting tool.
- It can help identify potential tool degradation.
- It is relevant to other failure mechanisms, particularly TWF and OSF.

Role in HDF:

Tool wear is not directly included in the AI4I HDF generation rule.

It may still be considered as an additional feature when building a machine learning model.

3.5 Engineering Interpretation of Heat Dissipation

Heat dissipation refers to the transfer of thermal energy from a machine or component to its surroundings.

Industrial machines generate heat during operation through several mechanisms.

3.5.1 Mechanical Friction

Moving mechanical components experience friction during operation.

Friction can convert mechanical energy into thermal energy, increasing component temperatures.

3.5.2 Electrical Losses

Electrical motors and associated components generate heat due to electrical resistance and other operating losses.

3.5.3 Cooling System Performance

Cooling systems help transfer heat away from machine components.

Depending on the equipment design, cooling may involve:

- Air circulation.
- Cooling fans.
- Heat exchangers.
- Liquid cooling.
- Lubrication and coolant circulation.

3.5.4 Environmental Temperature

Ambient temperature influences the thermal environment surrounding the machine.

A change in ambient temperature can affect the temperature gradient between machine components and their surroundings.

3.5.5 Operating Conditions

Machine speed and mechanical loading influence operating conditions and may affect heat generation and thermal behaviour.

Dataset limitation: These are general engineering explanations. AI4I does not contain direct measurements of cooling airflow, coolant pressure, fan condition, motor winding temperature or heat-exchanger performance.

Therefore, these factors should not be treated as individually verified causes of AI4I HDF labels.

3.6 Possible Engineering Symptoms

The AI4I dataset does not provide separate labels for physical cooling-system symptoms.

However, general industrial engineering knowledge identifies several possible indicators of thermal-management problems.

Possible symptom	Engineering interpretation
Elevated component temperature	Possible excessive heat generation or inadequate cooling
Reduced cooling airflow	Possible ventilation or fan-related problem
Abnormal temperature variation	Possible change in thermal operating conditions
Frequent thermal warnings	Possible recurring thermal-management issue
Cooling-system leakage	Possible issue in liquid-cooling equipment
Unexpected machine shutdown	Possible activation of equipment protection mechanisms

These symptoms are general engineering indicators and are not directly measured by the AI4I dataset.

3.7 Possible Contributing Factors

Several equipment-related conditions can affect heat dissipation.

3.7.1 Inadequate Ventilation

Restricted airflow may reduce the ability of a machine to release heat to its surroundings.

3.7.2 Cooling Fan Problems

A damaged or malfunctioning cooling fan may reduce airflow through the equipment.

3.7.3 Blocked Cooling Passages

Dust, debris or other obstructions may restrict airflow or coolant circulation.

3.7.4 Cooling System Inefficiency

Poor cooling-system performance may reduce the effectiveness of heat removal.

3.7.5 Excessive Operating Load

Increased mechanical or electrical loading can affect heat generation.

3.7.6 Environmental Conditions

High ambient temperature and unsuitable equipment placement may affect cooling performance.

Important: These are possible engineering contributing factors, not additional AI4I failure-generation conditions.

3.8 Inspection Procedures

The following inspection procedures are proposed as general guidance for industrial machinery.

Actual inspection must follow the machine manufacturer's instructions and applicable electrical and mechanical safety procedures.

3.8.1 Temperature Monitoring

Review the machine's available temperature measurements.

Inspect:

- Ambient temperature.
- Process temperature.
- Temperature difference.
- Historical temperature trends.

Compare measurements with equipment-specific operating limits.

3.8.2 Ventilation Inspection

Inspect ventilation openings and air passages for obstructions.

Check whether airflow is restricted by dust or debris.

3.8.3 Cooling Fan Inspection

Where cooling fans are installed:

- Inspect the fan for visible damage.
- Check for unusual noise.

- Verify that airflow is not obstructed.
- Follow manufacturer procedures for functional checks.

3.8.4 Cooling System Inspection

For equipment using liquid cooling or heat exchangers, inspect the system according to its manufacturer documentation.

Possible checks include:

- Coolant level.
- Leakage.
- Coolant circulation.
- Filter condition.
- Heat-exchanger cleanliness.

These checks are applicable only where the equipment includes the corresponding cooling components.

3.8.5 Operating Parameter Review

Review:

- Air temperature.
- Process temperature.
- Rotational speed.
- Torque.

Evaluate the measurements together rather than interpreting a single sensor reading in isolation.

3.9 Troubleshooting Procedures

The following table provides proposed troubleshooting guidance for suspected heat dissipation problems.

Observation	Possible explanation	Suggested action
Reduced temperature difference in AI4I	Dataset-defined HDF indicator may be present when RPM is also below 1380	Review both temperature measurements and rotational speed

Elevated component temperature	Possible cooling or operating-condition problem	Inspect the cooling system
Reduced airflow	Possible blocked ventilation	Inspect and clean ventilation passages according to equipment procedures
Cooling fan malfunction	Possible fan or electrical problem	Inspect the fan and associated components
Cooling system leakage	Possible loss of coolant	Follow manufacturer-approved leakage inspection and repair procedures
Repeated thermal warnings	Possible recurring thermal issue	Review historical temperature trends and investigate the cause

3.9.1 General Troubleshooting Workflow

Step 1: Identify the warning

Review the HDF prediction generated by the predictive maintenance system.

Step 2: Examine sensor measurements

Check air temperature, process temperature and rotational speed.

Step 3: Calculate temperature difference

Calculate the difference between process temperature and air temperature.

Step 4: Verify the dataset condition

Compare the temperature difference and rotational speed against the AI4I HDF thresholds.

Step 5: Inspect the cooling system

Where applicable, inspect ventilation, cooling fans and cooling equipment.

Step 6: Determine corrective action

Based on actual inspection results, take appropriate corrective action.

Step 7: Record maintenance activity

Document the measurements, inspection findings and corrective actions.

Step 8: Resume operation safely

Resume operation only after the equipment has been assessed as safe.

3.10 Preventive Maintenance Recommendations

Preventive maintenance helps reduce unexpected thermal problems and supports reliable equipment operation.

The following recommendations are proposed for general industrial applications.

3.10.1 Periodic Temperature Monitoring

Monitor operating temperatures and maintain historical records.

Identify unusual temperature trends rather than relying only on fixed thresholds.

3.10.2 Cooling System Maintenance

Inspect and maintain cooling equipment according to manufacturer recommendations.

3.10.3 Ventilation Maintenance

Keep ventilation openings and airflow passages free from obstructions.

3.10.4 Cooling Fan Inspection

Periodically inspect cooling fans for damage, unusual noise and operational problems.

3.10.5 Operating Condition Monitoring

Monitor rotational speed and torque to identify abnormal operating behaviour.

3.10.6 Maintenance Record Management

Maintain records containing:

- Date and time of inspection.
- Ambient temperature.
- Process temperature.
- Calculated temperature difference.
- Rotational speed.
- Cooling system condition.
- Observed abnormalities.
- Corrective actions.

3.10.7 Manufacturer-Recommended Operating Limits

Follow the manufacturer's recommended operating temperatures, speed limits and cooling-system maintenance intervals.

The AI4I HDF thresholds must not be substituted for actual equipment operating limits.

3.11 Relationship Between HDF and Other Failure Modes

HDF is one of five failure modes represented in AI4I.

It is important to understand that the individual failure indicators are not mutually exclusive.

Failure mode	Relationship with HDF
TWF	Represents a separate tool-wear mechanism
HDF	Associated with temperature difference and rotational speed
PWF	Associated with mechanical power conditions
OSF	Associated with torque, tool wear and product type
RNF	Generated independently of the operating parameters

A machine record can have more than one failure-mode label.

Consequently, the predictive maintenance system should support multiple failure predictions rather than treating all failure types as mutually exclusive categories.

3.12 Role of HDF in the Machine Learning System

HDF is treated as an individual binary classification target.

The model receives the available input features:

- Type.
- Air temperature.
- Process temperature.
- Rotational speed.
- Torque.
- Tool wear.

The model produces an HDF prediction and an associated probability estimate.

For example:

Output	Illustrative value
Predicted failure type	HDF
Predicted probability	85%

Air temperature	303 K
Process temperature	311 K
Temperature difference	8 K
Rotational speed	1300 rpm

These values are illustrative and do not represent an actual model prediction.

The dataset-defined HDF conditions are satisfied in this example because the temperature difference is below 8.6 K and rotational speed is below 1380 rpm.

The ML prediction is still a statistical model output and should not be confused with physical confirmation of a cooling-system defect.

3.13 Role of SHAP in HDF Diagnosis

SHAP (SHapley Additive exPlanations) explains how individual input features influence the HDF prediction generated by the trained machine learning model.

For HDF, SHAP may identify the contribution of:

- Air temperature.
- Process temperature.
- Rotational speed.
- Torque.
- Tool wear.
- Product type.

An illustrative explanation:

Feature	**Example SHAP contribution**	**Interpretation**
Temperature difference	+0.22	Pushes prediction towards HDF
Rotational speed	+0.18	Pushes prediction towards HDF
Torque	+0.03	Slight positive contribution
Tool wear	-0.04	Pushes prediction away from HDF

The temperature difference shown here is an engineered feature. If it is not included in the trained model, SHAP cannot directly report its contribution as an input feature.

SHAP values must be calculated from the actual trained model and input record.

Important: SHAP explains model behaviour. It does not establish that a cooling fan is damaged or that a heat exchanger is malfunctioning.

3.14 Role of RAG in HDF Diagnosis

The maintenance knowledge base contains technical information related to thermal management, cooling systems and HDF troubleshooting.

When the ML model predicts HDF, the diagnosis agent can retrieve relevant sections from this chapter.

Examples of useful retrieval topics:

- Heat dissipation failure definition.
- Temperature difference calculation.
- HDF generation condition.
- Cooling system inspection.
- Ventilation maintenance.
- Cooling fan troubleshooting.
- Thermal monitoring.
- Preventive maintenance procedures.

3.14.1 Example Internal Retrieval Query

Failure category: Heat Dissipation Failure (HDF)

Relevant parameters:

- Air temperature.
- Process temperature.
- Temperature difference.
- Rotational speed.

Retrieval request:

"Retrieve technical information about heat dissipation, temperature monitoring, cooling system inspection, ventilation problems, cooling fan maintenance and troubleshooting procedures for industrial machinery."

3.14.2 Example Retrieved Context

The RAG system may retrieve information explaining:

- How temperature difference is calculated.
- The AI4I-defined HDF conditions.
- Possible cooling-system inspection procedures.
- General thermal-management maintenance recommendations.

The retrieved context is then provided to the LLM along with the ML prediction and SHAP explanation.

3.15 Faithfulness Verification Requirements

The generated diagnosis must be supported by the retrieved technical evidence.

The faithfulness verifier should distinguish between dataset rules, engineering explanations and maintenance recommendations.

Claim type	Required supporting evidence
HDF temperature-difference threshold	AI4I dataset documentation
HDF rotational-speed threshold	AI4I dataset documentation
Temperature measurements	AI4I variable documentation
Cooling system explanation	Engineering reference
Ventilation inspection	Technical maintenance documentation
Cooling fan troubleshooting	Equipment maintenance documentation
ML prediction	Actual ML model output
Feature contributions	Actual SHAP output

The system must not make unsupported statements such as:

- A cooling fan is definitely damaged because HDF was predicted.
- The machine is physically overheating solely because the model predicts HDF.
- Every machine operating below 1380 rpm will experience HDF.
- An 8.6 K temperature difference is a universal industrial safety limit.

The model prediction should be treated as a warning that may require further investigation.

3.16 Knowledge Base Retrieval Metadata

The following metadata structure is recommended for storing HDF-related document chunks.

```
{
  "document_id": "AI4I_MANUAL_003",
  "chapter": 3,
  "failure_type": "HDF",
  "failure_name": "Heat Dissipation Failure",
  "section": "Cooling System Inspection",
  "equipment": "Simulated Industrial Milling Machine",
  "source_type": "Proposed Maintenance Guidance",
  "content": "Inspect ventilation openings, cooling fans "
        "and applicable cooling-system components "
        "according to manufacturer procedures."
}
```

Suggested section categories:

Category	Description
Failure_Definition	HDF explanation
Dataset_Rule	AI4I HDF generation conditions
Sensor_Parameters	Temperature and rotational-speed measurements
Temperature_Analysis	Temperature difference calculation
Engineering_Concepts	Heat generation and dissipation
Engineering_Symptoms	Possible thermal problem indicators
Inspection	Cooling and ventilation inspection
Troubleshooting	Corrective actions
Preventive_Maintenance	Preventive recommendations

Limitations	Dataset and prediction limitations

These categories can be used as metadata filters during vector retrieval.

3.17 Chapter Summary

Heat Dissipation Failure is a simulated failure mechanism in the AI4I 2020 Predictive Maintenance Dataset.

It is generated when two operating conditions occur together:

1. Temperature difference between process temperature and air temperature is below 8.6 K.
2. Rotational speed is below 1380 rpm.

Air temperature, process temperature and rotational speed are the primary measurements associated with the dataset-defined HDF condition.

The proposed maintenance procedures focus on temperature monitoring, cooling-system inspection, ventilation maintenance, cooling fan inspection and operating-condition monitoring.

Within the proposed research architecture:

1. The ML model estimates the likelihood of HDF.
2. SHAP explains the features influencing the prediction.
3. RAG retrieves relevant thermal-management knowledge and maintenance procedures.
4. The LLM generates a diagnosis and recommended actions.
5. The faithfulness verifier checks the generated claims against the retrieved evidence.

This integration connects thermal-condition predictions with external technical knowledge to support explainable and evidence-grounded predictive maintenance.

3.18 References

1. UCI Machine Learning Repository. (2020). *AI4I 2020 Predictive Maintenance Dataset*. https://doi.org/10.24432/C5HS5C
2. Matzka, S. (2020). *Explainable Artificial Intelligence for Predictive Maintenance Applications*. Third International Conference on Artificial Intelligence for Industries (AI4I).

3. ISO 13381-1:2015. *Condition monitoring and diagnostics of machines — Prognostics — Part 1: General guidelines*.
4. ISO 17359:2018. *Condition monitoring and diagnostics of machines — General guidelines*.

Source distinction: The HDF threshold conditions are based on AI4I dataset documentation. The cooling-system explanations, symptoms, inspection procedures and maintenance recommendations are general engineering knowledge and proposed guidance. Actual maintenance decisions must follow the relevant equipment manufacturer's specifications.

AI4I 2020 Predictive Maintenance Knowledge Manual

Chapter 4: Power Failure (PWF) – Technical Analysis and Maintenance Procedures

Manual: AI4I 2020 Predictive Maintenance Knowledge Manual
Failure Category: Power Failure (PWF)
Dataset: AI4I 2020 Predictive Maintenance Dataset
Application: Simulated Industrial Milling Machine

4.1 Introduction

Power Failure (PWF) is one of the five simulated failure mechanisms included in the AI4I 2020 Predictive Maintenance Dataset.

In industrial machinery, mechanical power describes the rate at which mechanical work is performed. It is influenced by the torque applied to a rotating component and its rotational speed.

An abnormal mechanical power condition may indicate that a machine is operating outside its intended mechanical operating range.

The AI4I dataset represents Power Failure using calculated mechanical power derived from two operating measurements:

1. Torque.
2. Rotational speed.

The dataset identifies PWF when calculated mechanical power falls below 3500 W or exceeds 9000 W.

This chapter explains the dataset-defined PWF generation mechanism, relevant sensor measurements, mechanical power calculations, possible engineering interpretations, inspection procedures, troubleshooting and preventive maintenance recommendations.

4.2 Definition of Power Failure

Power Failure in AI4I refers to an abnormal mechanical power condition identified using the relationship between torque and rotational speed.

Mechanical power describes how quickly mechanical work is transferred by a rotating system.

For a rotating machine, mechanical power depends on:

- Applied torque.

- Angular velocity.

If the calculated mechanical power is outside the defined operating range, the AI4I simulation can label the record as a Power Failure.

Important distinction: The AI4I PWF mechanism is based on calculated mechanical power. It does not directly measure electrical input power, voltage, current or power factor.

4.3 Dataset-Defined PWF Generation Mechanism

According to the AI4I 2020 dataset documentation, PWF is generated when mechanical power falls outside the specified range.

The dataset defines the following thresholds:

Parameter	Dataset condition
Lower mechanical power threshold	3500 W
Upper mechanical power threshold	9000 W
Failure condition 1	Mechanical power below 3500 W
Failure condition 2	Mechanical power above 9000 W

Therefore, the PWF condition can be represented as:

$$
PWF =
\begin{cases}
1, & P < 3500 \text{ W}\\
1, & P > 9000 \text{ W}\\
0, & 3500 \leq P \leq 9000 \text{ W}
\end{cases}
$$

Where (P) represents calculated mechanical power.

The two abnormal conditions are independent alternatives. Either condition can trigger the PWF label.

These thresholds are specific to the AI4I simulation and should not be treated as universal operating limits for industrial motors.

4.4 Mechanical Power Calculation

Mechanical power in a rotating system is calculated using torque and angular velocity.

The general formula is:

[
P = T \times \omega
]

Where:

- (P) = Mechanical power in watts (W).
- (T) = Torque in newton-metres (Nm).
- (\omega) = Angular velocity in radians per second (rad/s).

The AI4I dataset provides rotational speed in revolutions per minute (rpm), so the rotational speed must first be converted into angular velocity.

4.4.1 Conversion from RPM to Angular Velocity

One complete revolution corresponds to (2\pi) radians.

One minute contains 60 seconds.

Therefore:

[
\omega = \frac{2\pi \times RPM}{60}
]

Substituting this into the mechanical power equation:

[
P = T \times \frac{2\pi \times RPM}{60}
]

The final formula is:

[
\boxed{P = \frac{2\pi \times T \times RPM}{60}}
]

This formula is used to calculate mechanical power from the AI4I torque and rotational-speed measurements.

4.4.2 Example 1: Normal Mechanical Power

Assume:

- Torque = 40 Nm.
- Rotational speed = 1500 rpm.

Substituting into the formula:

$$
P = \frac{2\pi \times 40 \times 1500}{60}
$$

$$
P = 6283.19 \text{ W}
$$

The calculated mechanical power is approximately 6283.19 W.

Since this value is between 3500 W and 9000 W, it does not satisfy the AI4I PWF condition.

4.4.3 Example 2: Low Mechanical Power

Assume:

- Torque = 30 Nm.
- Rotational speed = 1000 rpm.

$$
P = \frac{2\pi \times 30 \times 1000}{60}
$$

$$
P = 3141.59 \text{ W}
$$

The calculated mechanical power is approximately 3141.59 W.

Since:

$$
3141.59 < 3500
$$

The record satisfies the low-power PWF condition.

4.4.4 Example 3: High Mechanical Power

Assume:

- Torque = 60 Nm.
- Rotational speed = 1500 rpm.

$$
P = \frac{2\pi \times 60 \times 1500}{60}
$$

$$
P = 9424.78 \text{ W}
$$

The calculated mechanical power is approximately 9424.78 W.

Since:

$$
9424.78 > 9000
$$

The record satisfies the high-power PWF condition.

4.4.5 Example Comparison

Scenario	Torque	RPM	Mechanical power	PWF condition
A	40 Nm	1500	6283.19 W	No
B	30 Nm	1000	3141.59 W	Low-power PWF
C	60 Nm	1500	9424.78 W	High-power PWF

These examples demonstrate why torque and rotational speed must be considered together when evaluating mechanical power.

4.5 Relevant Sensor Parameters

The following variables are important when analyzing PWF.

4.5.1 Torque [Nm]

Dataset column: Torque [Nm]

Unit: Newton-metre (Nm)

Torque represents the rotational force applied to the machine's rotating system.

Engineering significance:

- Torque represents the mechanical loading of the rotating system.
- Higher torque indicates greater rotational force.
- Torque contributes directly to mechanical power.
- Changes in torque can affect the machine's operating condition.

Role in PWF:

Torque is one of the two primary measurements used to calculate mechanical power.

4.5.2 Rotational Speed [rpm]

Dataset column: Rotational speed [rpm]

Unit: Revolutions per minute (rpm)

Rotational speed represents the number of revolutions performed by the rotating component per minute.

Engineering significance:

- It describes the operating speed of the rotating system.
- It influences mechanical power.
- It helps identify changes in machine operating conditions.
- It is useful for monitoring rotational behaviour.

Role in PWF:

Rotational speed is the second primary measurement used in the mechanical power calculation.

4.5.3 Air Temperature [K]

Dataset column: Air temperature [K]

Unit: Kelvin (K)

Air temperature represents the surrounding environmental temperature.

Role in PWF:

Air temperature is not directly used in the AI4I PWF generation formula. It can still provide additional operating context to a machine learning model.

4.5.4 Process Temperature [K]

Dataset column: Process temperature [K]

Unit: Kelvin (K)

Process temperature represents the operating temperature associated with the simulated manufacturing process.

Role in PWF:

Process temperature is not directly used in the AI4I PWF generation formula. However, it can provide supporting information about the thermal operating condition.

4.5.5 Tool Wear [min]

Dataset column: Tool wear [min]

Unit: Minutes

Tool wear represents accumulated cutting-tool usage.

Role in PWF:

Tool wear is not directly used in the AI4I PWF generation rule. It is relevant to other failure mechanisms, particularly TWF and OSF.

4.6 Engineering Interpretation of Mechanical Power

Mechanical power represents the rate at which mechanical work is performed.

In rotating machinery, power is influenced by both torque and rotational speed.

4.6.1 Low Mechanical Power

Low mechanical power may occur when torque, rotational speed or both are relatively low.

Possible engineering explanations include:

- Reduced mechanical loading.
- Reduced operating speed.
- Changes in the operating process.
- Drive-system or mechanical transmission problems.

However, low mechanical power does not automatically mean that a machine is defective. Some machines operate normally at low power depending on their operating conditions.

4.6.2 High Mechanical Power

High mechanical power may occur when torque, rotational speed or both are relatively high.

Possible engineering explanations include:

- Increased mechanical loading.
- Higher rotational speed.
- Increased process resistance.

- Abnormal operating conditions.

Actual equipment limits must be determined from the machine manufacturer's specifications.

4.6.3 Relationship Between Torque, RPM and Power

Mechanical power is directly proportional to torque and rotational speed.

For a fixed rotational speed:

- Increasing torque increases mechanical power.

For a fixed torque:

- Increasing rotational speed increases mechanical power.

Therefore, abnormal power conditions can result from changes in either parameter.

4.7 Possible Engineering Symptoms

The AI4I dataset does not contain separate physical symptoms or direct electrical measurements for PWF.

However, general industrial engineering knowledge identifies possible indicators of abnormal mechanical power conditions.

Possible symptom	Engineering interpretation
Reduced machine output	Possible reduction in delivered mechanical power
Abnormal rotational speed	Possible drive or operating-condition problem
Increased mechanical loading	Possible excessive process resistance
Unusual vibration	Possible mechanical or transmission abnormality
Abnormal motor temperature	Possible excessive loading or thermal issue
Repeated overload warnings	Possible recurring operating-condition problem

These symptoms are general engineering indicators and are not directly measured or individually labelled in AI4I.

4.8 Possible Contributing Factors

Mechanical power abnormalities may be associated with different machine and process conditions.

4.8.1 Excessive Mechanical Loading

An increase in the resistance experienced by the rotating system may increase the torque requirement.

4.8.2 Reduced Rotational Speed

Changes in rotational speed may affect the mechanical power delivered by the machine.

4.8.3 Drive-System Problems

Issues in the drive or transmission system may affect the torque and rotational speed delivered to the rotating components.

4.8.4 Unsuitable Operating Conditions

Operating outside the equipment's recommended speed and loading range may result in abnormal mechanical behaviour.

4.8.5 Mechanical Resistance

Unexpected friction or resistance in mechanical components may change the torque requirement.

Dataset limitation: AI4I does not directly identify these physical causes. They are general engineering possibilities and should not be interpreted as confirmed explanations for individual PWF records.

4.9 Inspection Procedures

The following inspection procedures are proposed as general guidance for machinery experiencing suspected abnormal mechanical power conditions.

Actual inspection must follow the machine manufacturer's instructions and applicable safety procedures.

4.9.1 Torque Measurement Review

Review the available torque measurements.

Check for:

- Unusual increases in torque.
- Unexpected reductions in torque.
- Sudden changes in torque behaviour.
- Deviations from historical operating conditions.

4.9.2 Rotational Speed Inspection

Review rotational-speed measurements.

Check for:

- Unexpected speed reduction.
- Unusual speed variation.
- Operating speed outside equipment specifications.
- Changes in rotational behaviour.

4.9.3 Mechanical Power Calculation

Calculate mechanical power using the measured torque and rotational speed.

Compare the calculated value against the AI4I thresholds when performing dataset analysis.

For real equipment, compare against the manufacturer's specified operating range.

4.9.4 Drive and Transmission Inspection

Where applicable, inspect the drive and transmission system for visible damage, abnormal noise or other irregularities.

4.9.5 Electrical System Inspection

For actual industrial equipment, qualified personnel may inspect the electrical supply, motor and drive system using suitable procedures and instruments.

The AI4I dataset does not provide voltage, current or power-factor measurements, so these cannot be evaluated directly from its records.

4.10 Troubleshooting Procedures

The following table provides proposed troubleshooting guidance for suspected PWF conditions.

Observation	Possible explanation	Suggested action
Calculated power below 3500 W	Low mechanical power under the AI4I rule	Review torque and rotational speed
Calculated power above 9000 W	High mechanical power under the AI4I rule	Review loading and rotational speed
Unexpected torque variation	Possible change in mechanical loading	Inspect the mechanical operating condition
Unexpected speed reduction	Possible drive or operating-condition problem	Review the drive system

Repeated abnormal power warnings	Possible recurring operating issue	Review historical power trends
Unusual vibration or noise	Possible mechanical abnormality	Conduct an appropriate mechanical inspection

4.10.1 General Troubleshooting Workflow

Step 1: Identify the warning

Review the PWF prediction generated by the predictive maintenance system.

Step 2: Examine sensor measurements

Check torque and rotational speed.

Step 3: Calculate mechanical power

Use:

[
P = \frac{2\pi \times T \times RPM}{60}
]

Step 4: Verify the dataset condition

Check whether the calculated power is below 3500 W or above 9000 W.

Step 5: Investigate operating conditions

Review mechanical loading, speed and relevant operating history.

Step 6: Inspect the equipment

Where applicable, inspect the drive, transmission and mechanical components according to manufacturer procedures.

Step 7: Determine corrective action

Take corrective action based on the actual inspection findings.

Step 8: Record maintenance activity

Document the measurements, observations, inspection findings and corrective actions.

Step 9: Resume operation safely

Resume operation only after the equipment has been assessed as safe.

4.11 Preventive Maintenance Recommendations

Preventive maintenance helps identify abnormal mechanical operating conditions before they lead to unexpected equipment interruptions.

The following recommendations are proposed for general industrial applications.

4.11.1 Continuous Torque Monitoring

Monitor torque measurements to identify abnormal changes in mechanical loading.

4.11.2 Rotational Speed Monitoring

Monitor rotational speed and compare it with equipment-specific operating limits.

4.11.3 Mechanical Power Monitoring

Calculate mechanical power using torque and rotational speed.

Maintain historical records to identify unusual operating trends.

4.11.4 Drive-System Maintenance

Inspect drive and transmission components according to the equipment manufacturer's maintenance schedule.

4.11.5 Mechanical Component Inspection

Inspect relevant rotating and transmission components for abnormal noise, wear, damage or vibration.

4.11.6 Electrical System Maintenance

Follow manufacturer-recommended inspection procedures for motors, drives and electrical connections.

4.11.7 Maintenance Record Management

Maintain records containing:

- Date and time of inspection.
- Torque measurements.
- Rotational speed.
- Calculated mechanical power.
- Operating conditions.
- Inspection findings.
- Corrective actions.
- Maintenance history.

4.11.8 Manufacturer-Recommended Operating Limits

Always follow equipment-specific torque, speed and power specifications.

The AI4I thresholds of 3500 W and 9000 W are dataset-specific values and should not be applied directly to real industrial machines.

4.12 Relationship Between PWF and Other Failure Modes

PWF is one of the five individual failure indicators in AI4I.

The relationship between failure modes is summarized below.

Failure mode	Relationship with PWF
TWF	Associated with the simulated tool-wear threshold
HDF	Associated with temperature difference and rotational speed
PWF	Associated with calculated mechanical power
OSF	Associated with torque, tool wear and product type
RNF	Generated independently of operating parameters

PWF and OSF both involve torque-related operating information, but their failure-generation conditions are different.

PWF uses calculated mechanical power, whereas OSF uses the product of torque and tool wear with product-specific thresholds.

A record can be associated with multiple failure labels.

Therefore, the predictive maintenance system should support separate binary predictions for individual failure modes.

4.13 Role of PWF in the Machine Learning System

PWF is treated as an individual binary classification target.

The machine learning model receives the available input features:

- Type.
- Air temperature.
- Process temperature.
- Rotational speed.
- Torque.

- Tool wear.

The model generates:

- PWF prediction.
- PWF probability estimate.

An additional engineered feature can be calculated:

Mechanical_power_W = Torque * Rotational_speed * 2 * pi / 60

This engineered feature represents the mechanical power calculation used by the dataset's PWF generation rule.

4.13.1 Illustrative Prediction

Output	Illustrative value
Predicted failure type	PWF
Predicted probability	87%
Torque	60 Nm
Rotational speed	1500 rpm
Calculated mechanical power	9424.78 W

The values are illustrative and do not represent an actual trained-model prediction.

The calculated mechanical power exceeds 9000 W, satisfying the high-power condition in the AI4I PWF rule.

4.14 Role of SHAP in PWF Diagnosis

SHAP (SHapley Additive exPlanations) explains how individual input features contribute to the PWF prediction generated by the trained model.

For PWF, SHAP may identify the contributions of:

- Torque.
- Rotational speed.
- Engineered mechanical power, if included in the model.
- Product type.
- Air temperature.
- Process temperature.

- Tool wear.

An illustrative explanation:

Feature	Example SHAP contribution	Interpretation
Mechanical power	+0.28	Pushes prediction towards PWF
Torque	+0.14	Pushes prediction towards PWF
Rotational speed	+0.09	Pushes prediction towards PWF
Tool wear	-0.03	Pushes prediction away from PWF

These values are illustrative and not actual SHAP results.

Important: If mechanical power is engineered from torque and rotational speed and all three are included in the model, their SHAP contributions may overlap because the features are mathematically related.

SHAP explains the model's behaviour. It does not prove an electrical fault or identify the physical component responsible for abnormal power.

4.15 Role of RAG in PWF Diagnosis

The maintenance knowledge base contains technical information about mechanical power, torque, rotational speed, drive systems and troubleshooting procedures.

When the ML model predicts PWF, the diagnosis agent can retrieve relevant sections from this chapter.

Examples of useful retrieval topics include:

- Power Failure definition.
- Mechanical power calculation.
- Torque and rotational-speed relationship.
- Low mechanical power.
- High mechanical power.
- Drive-system inspection.
- Mechanical loading.
- Preventive maintenance procedures.

4.15.1 Example Internal Retrieval Query

Failure category: Power Failure (PWF)

Relevant parameters:

- Torque.
- Rotational speed.
- Calculated mechanical power.

Retrieval request:

"Retrieve technical information about abnormal mechanical power, torque and rotational-speed relationships, drive-system inspection, mechanical loading and preventive maintenance procedures for industrial rotating machinery."

4.15.2 Example Retrieved Context

The RAG system may retrieve information explaining:

- How mechanical power is calculated.
- The AI4I-defined PWF thresholds.
- Possible explanations for abnormal mechanical power.
- General inspection procedures for drive and transmission systems.
- Recommended monitoring practices.

The retrieved context is then provided to the LLM together with the ML prediction and SHAP explanation.

4.16 Faithfulness Verification Requirements

The generated diagnosis must be supported by retrieved technical evidence.

The faithfulness verifier should distinguish dataset-specific rules from general electrical and mechanical engineering knowledge.

Claim type	Required supporting evidence
PWF lower threshold	AI4I dataset documentation
PWF upper threshold	AI4I dataset documentation
Mechanical power formula	Mechanical engineering reference
Torque measurement	AI4I variable documentation
Rotational speed measurement	AI4I variable documentation
Drive-system inspection	Equipment maintenance documentation

Electrical fault diagnosis	Relevant electrical maintenance documentation
ML prediction	Actual ML model output
Feature contribution	Actual SHAP output

The system must not generate unsupported statements such as:

- The electrical motor is definitely damaged because PWF is predicted.
- The machine is consuming excessive electrical energy because calculated mechanical power is high.
- Every machine operating above 9000 W will experience a power failure.
- The AI4I power thresholds represent universal motor operating limits.

A PWF prediction is an indication of a simulated abnormal mechanical power condition, not proof of a specific physical electrical fault.

4.17 Knowledge Base Retrieval Metadata

The following metadata structure is recommended for storing PWF-related document chunks.

```
{
  "document_id": "AI4I_MANUAL_004",
  "chapter": 4,
  "failure_type": "PWF",
  "failure_name": "Power Failure",
  "section": "Mechanical Power Calculation",
  "equipment": "Simulated Industrial Milling Machine",
  "source_type": "Dataset_Documentation",
  "content": "Mechanical power is calculated using "
       "torque and rotational speed. The formula "
       "is P = (2*pi*T*RPM)/60."
}
```

Suggested section categories:

Category	**Description**

Failure_Definition	PWF explanation
Dataset_Rule	AI4I power thresholds
Power_Calculation	Mechanical power formula
Sensor_Parameters	Torque and rotational speed
Engineering_Concepts	Mechanical power and loading
Engineering_Symptoms	Possible abnormal operating indicators
Inspection	Mechanical and drive-system inspection
Troubleshooting	Corrective actions
Preventive_Maintenance	Preventive recommendations
Limitations	Dataset and prediction limitations

These categories can be used as metadata filters during vector retrieval.

4.18 Chapter Summary

Power Failure is a simulated failure mechanism in the AI4I 2020 Predictive Maintenance Dataset.

It is identified using mechanical power calculated from torque and rotational speed.

The dataset defines two abnormal operating conditions:

1. Mechanical power below 3500 W.
2. Mechanical power above 9000 W.

The mechanical power calculation is:

[
P = \frac{2\pi \times T \times RPM}{60}
]

The proposed maintenance procedures focus on torque monitoring, rotational-speed monitoring, mechanical power analysis, drive-system inspection and preventive maintenance.

Within the proposed research architecture:

1. The ML model estimates the likelihood of PWF.
2. SHAP explains the features influencing the prediction.

3. RAG retrieves relevant mechanical power knowledge and maintenance procedures.
4. The LLM generates a diagnosis and recommended actions.
5. The faithfulness verifier checks the generated claims against the retrieved evidence.

This integration connects mechanical power predictions with technical knowledge to support explainable and evidence-grounded predictive maintenance.

4.19 References

1. UCI Machine Learning Repository. (2020). *AI4I 2020 Predictive Maintenance Dataset*. https://doi.org/10.24432/C5HS5C
2. Matzka, S. (2020). *Explainable Artificial Intelligence for Predictive Maintenance Applications*. Third International Conference on Artificial Intelligence for Industries (AI4I).
3. ISO 13381-1:2015. *Condition monitoring and diagnostics of machines — Prognostics — Part 1: General guidelines*.
4. ISO 17359:2018. *Condition monitoring and diagnostics of machines — General guidelines*.

Source distinction: The PWF thresholds and dataset variables are based on AI4I documentation. The mechanical power equation follows standard rotational mechanics. The engineering symptoms, inspection procedures and maintenance recommendations are general engineering knowledge and proposed guidance. Actual maintenance decisions must follow the relevant equipment manufacturer's specifications.

AI4I 2020 Predictive Maintenance Knowledge Manual

Chapter 5: Overstrain Failure (OSF) – Technical Analysis and Maintenance Procedures

Manual: AI4I 2020 Predictive Maintenance Knowledge Manual
Failure Category: Overstrain Failure (OSF)
Dataset: AI4I 2020 Predictive Maintenance Dataset
Application: Simulated Industrial Milling Machine

5.1 Introduction

Overstrain Failure (OSF) is one of the five simulated failure mechanisms included in the AI4I 2020 Predictive Maintenance Dataset.

In industrial machinery, mechanical components experience forces and stresses during operation. When mechanical loading becomes excessive, components may experience increased stress, deformation, accelerated wear or mechanical damage.

In machining operations, the combination of applied torque and accumulated tool wear can influence the mechanical loading conditions experienced by the cutting system.

The AI4I dataset represents Overstrain Failure using a combination of:

1. **Applied torque.**
2. **Accumulated tool wear.**
3. **Product type.**

Unlike Tool Wear Failure, which is primarily associated with a tool-wear threshold, OSF uses the product of torque and tool wear compared against product-specific thresholds.

This chapter explains the dataset-defined OSF mechanism, engineering interpretation, relevant sensor measurements, possible symptoms, inspection procedures, troubleshooting and preventive maintenance recommendations.

5.2 Definition of Overstrain Failure

Overstrain Failure refers to a simulated failure condition associated with excessive mechanical loading.

In the AI4I dataset, OSF is generated when the product of torque and tool wear exceeds a threshold determined by the product type.

The failure mechanism combines:

- **Torque, representing applied rotational force.**
- **Tool wear, representing accumulated cutting-tool usage.**
- **Product type, determining the applicable failure threshold.**

The dataset uses these variables to represent a mechanical overstrain condition.

Important: The AI4I OSF rule is a synthetic failure-generation mechanism. It is not a direct measurement of material stress, strain, fatigue damage or physical component deformation.

5.3 Dataset-Defined OSF Generation Mechanism

According to the AI4I 2020 dataset documentation, OSF is generated when the product of torque and tool wear exceeds a product-specific threshold.

5.3.1 OSF Calculation

The OSF calculation is:

[
OSF_{\text{value}} = T \times W
]

Where:

- **(T) = Torque in Newton-metres (Nm).**
- **(W) = Tool wear in minutes.**
- **(OSF_{\text{value}}) = Torque-wear product.**

The resulting numerical value is compared against the threshold associated with the product type.

5.3.2 Product-Specific Thresholds

The AI4I dataset defines the following thresholds:

Product type	Product-specific threshold
L	11,000
M	12,000
H	13,000

These thresholds are applied to the product of torque and tool wear.

5.3.3 OSF Failure Condition

The OSF rule can be represented as:

$$
\begin{cases}
1, & T \times W > 11000,\quad Type=L\\
1, & T \times W > 12000,\quad Type=M\\
1, & T \times W > 13000,\quad Type=H\\
0, & \text{otherwise}
\end{cases}
$$

The failure condition is evaluated using the threshold associated with the corresponding product type.

5.3.4 Example 1: L-Type Product

Assume:

- **Product type = L.**
- **Torque = 60 Nm.**
- **Tool wear = 190 minutes.**

Calculation:

$$
OSF_{\text{value}} = 60 \times 190
$$

$$
OSF_{\text{value}} = 11400
$$

The L-type threshold is 11,000.

Since:

$$
11400 > 11000
$$

The record satisfies the OSF condition.

5.3.5 Example 2: M-Type Product

Assume:

- **Product type = M.**
- **Torque = 55 Nm.**
- **Tool wear = 210 minutes.**

Calculation:

$$
\mathbf{OSF_{\text{value}} = 55 \times 210}
$$

$$
\mathbf{OSF_{\text{value}} = 11550}
$$

The M-type threshold is 12,000.

Since:

$$
\mathbf{11550 < 12000}
$$

The record does not satisfy the OSF condition.

5.3.6 Example 3: H-Type Product

Assume:

- **Product type = H.**
- **Torque = 65 Nm.**
- **Tool wear = 210 minutes.**

Calculation:

$$
\mathbf{OSF_{\text{value}} = 65 \times 210}
$$

$$
\mathbf{OSF_{\text{value}} = 13650}
$$

The H-type threshold is 13,000.

Since:

[
13650 > 13000
]

The record satisfies the OSF condition.

5.3.7 Example Comparison

Scenario	Type	Torque	Tool wear	Product	OSF condition
A	L	60 Nm	190 min	11,400	Satisfied
B	M	55 Nm	210 min	11,550	Not satisfied
C	H	65 Nm	210 min	13,650	Satisfied

These examples demonstrate why the product type must be considered when evaluating OSF.

Note: The examples illustrate the dataset's threshold calculation. They do not establish universal industrial mechanical-load limits.

5.4 Relevant Sensor Parameters

The following variables are important when analyzing OSF.

5.4.1 Torque [Nm]

Dataset column: Torque [Nm]

Unit: Newton-metre (Nm)

Torque represents the rotational force applied to the machine's rotating system.

Engineering significance:

- **It indicates the mechanical loading of the rotating system.**
- **Higher torque represents greater rotational force.**
- **It contributes to mechanical power.**
- **Changes in torque may indicate variations in operating conditions.**

Role in OSF:

Torque is one of the two primary variables used in the AI4I OSF generation mechanism.

5.4.2 Tool Wear [min]

Dataset column: Tool wear [min]

Unit: Minutes

Tool wear represents accumulated cutting-tool usage time.

Engineering significance:

- **It provides information about tool usage history.**
- **It can indicate increasing exposure to wear.**
- **Excessive tool wear may affect machining performance.**
- **Tool condition can influence cutting resistance.**

Role in OSF:

Tool wear is the second primary variable used in the OSF calculation.

It is multiplied by torque to determine whether the product-specific threshold is exceeded.

5.4.3 Product Type

Dataset column: Type

Possible values:

- **L: Low-quality variant.**
- **M: Medium-quality variant.**
- **H: High-quality variant.**

Role in OSF:

Product type determines the threshold used in the OSF generation rule.

The thresholds are:

- **L: 11,000.**
- **M: 12,000.**
- **H: 13,000.**

Therefore, the same torque-wear product can result in different OSF labels depending on the product type.

5.4.4 Rotational Speed [rpm]

Dataset column: Rotational speed [rpm]

Unit: Revolutions per minute (rpm)

Rotational speed represents the operating speed of the rotating system.

Engineering significance:

- **It describes the rotational behaviour of the machine.**
- **It influences mechanical power.**
- **It provides additional operating context.**

Role in OSF:

Rotational speed is not directly included in the AI4I OSF generation formula. It can still provide additional information to the machine learning model.

5.4.5 Temperature Measurements

The dataset contains:

- **Air temperature [K].**
- **Process temperature [K].**

These measurements describe the thermal operating conditions.

Role in OSF:

Temperature measurements are not directly used in the AI4I OSF generation rule. They can be considered as additional input features during machine learning.

5.5 Engineering Interpretation of Mechanical Overstrain

Mechanical overstrain occurs when mechanical loading places a component under excessive stress relative to its design or operating limits.

The relationship between torque and mechanical stress depends on component geometry, material properties and loading conditions.

5.5.1 Mechanical Loading

Mechanical loading represents the forces acting on machine components during operation.

In rotating machinery, torque produces torsional loading on shafts and related components.

5.5.2 Tool Wear and Cutting Resistance

As a cutting tool degrades, its machining performance may change.

Depending on the material and cutting conditions, tool degradation may contribute to changes in cutting resistance and mechanical loading.

5.5.3 Combined Operating Effects

Torque and tool wear provide two different forms of operating information.

The AI4I OSF mechanism combines them through a mathematical product.

This represents a simplified simulation of an overstrain condition.

5.5.4 Product-Specific Behaviour

Different product types use different OSF thresholds.

The model must therefore consider product type when evaluating the relationship between torque and tool wear.

Dataset limitation: The torque-wear product is a synthetic indicator used by AI4I. It is not a direct physical measurement of stress or strain in a machine component.

5.6 Possible Engineering Symptoms

AI4I does not contain direct measurements of mechanical deformation or physical component damage.

However, general engineering knowledge identifies possible indicators of excessive mechanical loading.

Possible symptom	Engineering interpretation
Excessive vibration	Possible mechanical imbalance or loading abnormality
Unusual mechanical noise	Possible component wear or mechanical interference
Increased torque requirement	Possible increase in mechanical resistance
Poor machining quality	Possible cutting-tool degradation or process instability
Visible component deformation	Possible excessive mechanical loading
Repeated mechanical warnings	Possible recurring operating-condition problem

These symptoms are general engineering indicators and are not directly measured or individually labelled in AI4I.

5.7 Possible Contributing Factors

Mechanical overstrain can be influenced by several operating and equipment-related factors.

5.7.1 Excessive Mechanical Loading

Operating a machine beyond its recommended loading conditions may increase mechanical stress.

5.7.2 Tool Degradation

Worn or damaged cutting tools may affect machining resistance and process stability.

5.7.3 Unsuitable Machining Parameters

Unsuitable cutting speed, feed rate or cutting depth may influence mechanical loading.

5.7.4 Mechanical Misalignment

Misalignment in rotating or transmission components may contribute to abnormal mechanical behaviour.

5.7.5 Inadequate Lubrication

Insufficient lubrication may increase friction between mechanical components.

5.7.6 Improper Operating Conditions

Operating outside equipment-specific specifications may contribute to excessive mechanical stress.

Important: These are general engineering possibilities. AI4I does not identify them as the actual causes of individual OSF records.

5.8 Inspection Procedures

The following inspection procedures are proposed as general guidance for suspected mechanical overstrain conditions.

Actual inspection must follow the machine manufacturer's instructions and relevant safety procedures.

5.8.1 Torque Measurement Review

Review the torque measurements and identify unusual operating behaviour.

Check for:

- **Unexpected torque increases.**
- **Repeated high-torque readings.**

- **Sudden changes in torque.**
- **Deviations from historical measurements.**

5.8.2 Tool Condition Inspection

Inspect the cutting tool for:

- **Excessive wear.**
- **Cutting-edge damage.**
- **Chipping.**
- **Cracks.**
- **Deformation.**

5.8.3 Mechanical Component Inspection

Where applicable, inspect relevant mechanical components for visible damage, unusual wear, misalignment or abnormal movement.

5.8.4 Operating Parameter Review

Review:

- **Torque.**
- **Tool wear.**
- **Rotational speed.**
- **Product type.**
- **Process temperature.**

Compare measurements with the equipment manufacturer's operating specifications.

5.8.5 Mechanical Loading Assessment

Evaluate whether the machine is operating within its recommended loading conditions.

Actual mechanical stress assessment requires suitable equipment specifications and physical measurements; the AI4I torque-wear product alone is insufficient.

5.9 Troubleshooting Procedures

The following table provides proposed troubleshooting guidance for suspected OSF conditions.

Observation	Possible explanation	Suggested action
High torque-wear product	AI4I OSF threshold may be exceeded	Review torque, tool wear and product type
Excessive tool wear	Possible cutting-tool degradation	Inspect the tool and assess replacement
Repeated high torque	Possible increased mechanical resistance	Inspect the mechanical operating condition
Abnormal vibration	Possible mechanical instability	Conduct an appropriate mechanical inspection
Poor machining quality	Possible tool degradation or process issue	Inspect tooling and machining parameters
Repeated OSF warnings	Possible recurring loading problem	Review operating history and maintenance records

5.9.1 General Troubleshooting Workflow

Step 1: Identify the warning

Review the OSF prediction generated by the predictive maintenance system.

Step 2: Examine sensor measurements

Check torque, tool wear and product type.

Step 3: Calculate the torque-wear product

Use:

[
OSF_{\text{value}} = T \times W
]

Step 4: Identify the applicable threshold

Determine the product-specific threshold:

- **L: 11,000.**
- **M: 12,000.**
- **H: 13,000.**

Step 5: Verify the dataset condition

Compare the calculated product with the appropriate threshold.

Step 6: Inspect the tool and mechanical system

Where applicable, inspect the cutting tool and relevant mechanical components.

Step 7: Determine corrective action

Take corrective action based on physical inspection and equipment specifications.

Step 8: Record maintenance activity

Document measurements, inspection findings and corrective actions.

Step 9: Resume operation safely

Resume operation only after the equipment has been assessed as safe.

5.10 Preventive Maintenance Recommendations

Preventive maintenance helps reduce excessive mechanical loading and supports reliable machining operations.

The following recommendations are proposed for general industrial applications.

5.10.1 Torque Monitoring

Monitor torque measurements to identify abnormal changes in mechanical loading.

5.10.2 Tool Wear Monitoring

Track accumulated tool usage and inspect cutting tools according to manufacturer recommendations.

5.10.3 Condition-Based Tool Replacement

Consider tool replacement based on physical condition, machining performance and equipment-specific acceptance criteria.

5.10.4 Operating Parameter Optimization

Follow appropriate cutting speeds, feed rates and cutting depths for the relevant tool and material combination.

5.10.5 Mechanical Component Inspection

Inspect relevant mechanical components for abnormal vibration, wear, damage and misalignment.

5.10.6 Lubrication Maintenance

Follow equipment-specific lubrication requirements and recommended maintenance intervals.

5.10.7 Maintenance Record Management

Maintain records containing:

- **Date and time of inspection.**
- **Torque measurements.**
- **Accumulated tool wear.**
- **Product type.**
- **Calculated torque-wear product.**
- **Tool inspection results.**
- **Mechanical inspection findings.**
- **Corrective maintenance actions.**

5.10.8 Manufacturer-Recommended Operating Limits

Follow manufacturer specifications for torque, mechanical loading, tooling and machine operation.

The AI4I product-specific OSF thresholds must not be applied as universal industrial mechanical-stress limits.

5.11 Relationship Between OSF and Other Failure Modes

OSF is one of five individual failure indicators in the AI4I dataset.

Its relationship with other failure mechanisms is summarized below.

Failure mode	Relationship with OSF
TWF	Both involve tool wear, but use different failure-generation mechanisms
HDF	Uses temperature difference and rotational speed
PWF	Uses calculated mechanical power
OSF	Uses torque, tool wear and product-specific thresholds
RNF	Generated independently of operating parameters

5.11.1 Relationship Between TWF and OSF

TWF and OSF both involve tool wear, but their conditions are different.

TWF:

Uses a randomly selected tool-wear threshold between 200 and 240 minutes in the simulated failure-generation process.

OSF:

Uses the product of torque and tool wear compared against product-specific thresholds.

Therefore, a record can satisfy both TWF and OSF conditions.

The predictive maintenance system should support independent binary predictions for these failure modes.

5.11.2 Relationship Between PWF and OSF

PWF and OSF both involve torque, but their mathematical conditions are different.

- **PWF uses torque multiplied by rotational speed to calculate mechanical power.**
- **OSF uses torque multiplied by accumulated tool wear.**

This distinction is important when interpreting ML predictions and SHAP explanations.

5.12 Role of OSF in the Machine Learning System

OSF is treated as an individual binary classification target.

The machine learning model receives the available input features:

- **Type.**
- **Air temperature.**
- **Process temperature.**
- **Rotational speed.**
- **Torque.**
- **Tool wear.**

The model produces:

- **OSF prediction.**
- **OSF probability estimate.**

An additional engineered feature can be calculated:

Torque_wear_interaction = Torque * Tool_wear

This feature represents the mathematical interaction used in the AI4I OSF generation mechanism.

5.12.1 Illustrative Prediction

Output	Illustrative value
Predicted failure type	OSF
Predicted probability	89%
Product type	L
Torque	60 Nm
Tool wear	190 minutes
Torque-wear product	11,400
Applicable threshold	11,000

The values are illustrative and do not represent an actual model prediction.

Since 11,400 exceeds the L-type threshold of 11,000, the example satisfies the dataset-defined OSF condition.

5.13 Role of SHAP in OSF Diagnosis

SHAP (SHapley Additive exPlanations) explains how individual input features influence the OSF prediction generated by the trained machine learning model.

For OSF, SHAP may identify contributions from:

- **Torque.**
- **Tool wear.**
- **Product type.**
- **Torque-wear interaction, if included in the model.**
- **Rotational speed.**
- **Temperature measurements.**

An illustrative explanation:

Feature	Example SHAP contribution	Interpretation

Torque-wear interaction	**+0.27**	**Pushes prediction towards OSF**
Torque	**+0.13**	**Pushes prediction towards OSF**
Tool wear	**+0.16**	**Pushes prediction towards OSF**
Product type	**+0.05**	**Pushes prediction towards OSF**
Rotational speed	**-0.03**	**Pushes prediction away from OSF**

These values are illustrative and are not actual SHAP results.

Important: If torque-wear interaction is engineered from torque and tool wear, SHAP contributions can overlap because these features are mathematically related.

SHAP explains the model's behaviour. It does not directly measure physical mechanical stress or prove component deformation.

5.14 Role of RAG in OSF Diagnosis

The maintenance knowledge base contains technical information related to mechanical loading, tool wear, inspection procedures and preventive maintenance.

When the ML model predicts OSF, the diagnosis agent can retrieve relevant sections from this chapter.

Examples of useful retrieval topics include:

- **Overstrain Failure definition.**
- **Torque-wear interaction.**
- **Product-specific OSF thresholds.**
- **Mechanical loading.**
- **Tool condition inspection.**
- **Mechanical component inspection.**
- **Troubleshooting procedures.**
- **Preventive maintenance recommendations.**

5.14.1 Example Internal Retrieval Query

Failure category: Overstrain Failure (OSF)

Relevant parameters:

- **Torque.**
- **Tool wear.**
- **Product type.**
- **Torque-wear interaction.**

Retrieval request:

"Retrieve technical information about mechanical overstrain, torque and tool-wear relationships, product-specific AI4I failure thresholds, cutting-tool inspection, mechanical loading and preventive maintenance procedures for industrial milling machinery."

5.14.2 Example Retrieved Context

The RAG system may retrieve information explaining:

- **The AI4I OSF calculation.**
- **Product-specific failure thresholds.**
- **The relationship between torque and tool wear.**
- **Possible engineering symptoms of excessive mechanical loading.**
- **General inspection and maintenance procedures.**

The retrieved information is provided to the LLM together with the ML prediction and SHAP explanation.

5.15 Faithfulness Verification Requirements

The generated diagnosis must be supported by the retrieved technical evidence.

The faithfulness verifier should distinguish between dataset-defined conditions, general engineering knowledge and proposed maintenance recommendations.

Claim type	Required supporting evidence
OSF definition	AI4I dataset documentation
Torque-wear product	AI4I dataset documentation
Product-specific thresholds	AI4I dataset documentation
Torque measurement	AI4I variable documentation

Tool-wear measurement	**AI4I variable documentation**
Mechanical stress explanation	**Engineering reference**
Tool inspection	**Technical maintenance documentation**
Mechanical troubleshooting	**Equipment maintenance documentation**
ML prediction	**Actual ML model output**
Feature contribution	**Actual SHAP output**

The system must not generate unsupported claims such as:

- **A machine shaft is definitely damaged because OSF is predicted.**
- **The machine has exceeded its physical stress limit solely because the torque-wear product is high.**
- **Every machine using an L-type product has an 11,000 mechanical-stress limit.**
- **A tool must be replaced solely because the model predicts OSF.**

The OSF prediction represents a simulated failure classification. Physical inspection and equipment-specific specifications are necessary before making actual maintenance decisions.

5.16 Knowledge Base Retrieval Metadata

The following metadata structure is recommended for storing OSF-related document chunks.

```
{
  "document_id": "AI4I_MANUAL_005",
  "chapter": 5,
  "failure_type": "OSF",
  "failure_name": "Overstrain Failure",
  "section": "OSF Generation Mechanism",
  "equipment": "Simulated Industrial Milling Machine",
  "source_type": "Dataset_Documentation",
  "content": "Overstrain Failure is generated when "
      "the product of torque and tool wear "
```

"exceeds the product-specific threshold."

}

Suggested section categories:

Category	Description
Failure_Definition	OSF explanation
Dataset_Rule	Product-specific OSF thresholds
OSF_Calculation	Torque-wear product
Sensor_Parameters	Torque, tool wear and product type
Engineering_Concepts	Mechanical loading and overstrain
Engineering_Symptoms	Possible mechanical problem indicators
Inspection	Tool and mechanical inspection
Troubleshooting	Corrective actions
Preventive_Maintenance	Preventive recommendations
Limitations	Dataset and prediction limitations

These categories can be used as metadata filters during vector retrieval.

5.17 Chapter Summary

Overstrain Failure is a simulated failure mechanism in the AI4I 2020 Predictive Maintenance Dataset.

It is generated when the product of torque and tool wear exceeds a product-specific threshold.

The thresholds are:

- **L: 11,000.**
- **M: 12,000.**
- **H: 13,000.**

Torque, tool wear and product type are the primary variables associated with the dataset-defined OSF condition.

The proposed maintenance procedures focus on torque monitoring, tool inspection, mechanical component assessment, operating-condition monitoring and preventive maintenance.

Within the proposed research architecture:

1. **The ML model estimates the likelihood of OSF.**
2. **SHAP explains the features influencing the prediction.**
3. **RAG retrieves relevant mechanical loading knowledge and maintenance procedures.**
4. **The LLM generates a diagnosis and recommended actions.**
5. **The faithfulness verifier checks the generated claims against the retrieved evidence.**

This integration connects torque-wear-based failure predictions with external engineering knowledge to support explainable and evidence-grounded predictive maintenance.

5.18 References

1. **UCI Machine Learning Repository. (2020). *AI4I 2020 Predictive Maintenance Dataset*. https://doi.org/10.24432/C5HS5C**
2. **Matzka, S. (2020). *Explainable Artificial Intelligence for Predictive Maintenance Applications*. Third International Conference on Artificial Intelligence for Industries (AI4I).**
3. **ISO 13381-1:2015. *Condition monitoring and diagnostics of machines — Prognostics — Part 1: General guidelines*.**
4. **ISO 17359:2018. *Condition monitoring and diagnostics of machines — General guidelines*.**

Source distinction: The torque-wear calculation and product-specific thresholds are based on AI4I dataset documentation. The engineering explanations, symptoms, inspection procedures and maintenance recommendations are general engineering knowledge and proposed guidance. Actual maintenance decisions must follow the relevant equipment manufacturer's specifications.

AI4I 2020 Predictive Maintenance Knowledge Manual

Chapter 6: Random Failure (RNF) – Technical Analysis and Maintenance Procedures

Manual: AI4I 2020 Predictive Maintenance Knowledge Manual
Failure Category: Random Failure (RNF)
Dataset: AI4I 2020 Predictive Maintenance Dataset
Application: Simulated Industrial Milling Machine

6.1 Introduction

Random Failure (RNF) is one of the five simulated failure mechanisms included in the AI4I 2020 Predictive Maintenance Dataset.

In industrial environments, equipment failures may occur due to unexpected events that are difficult to predict using ordinary operating measurements.

Examples of real-world unexpected events may include sudden component damage, unexpected electrical interruptions, accidental mechanical damage or other unforeseen operating disturbances.

The AI4I dataset represents Random Failure through a randomly generated failure event that is independent of the available process measurements.

Unlike TWF, HDF, PWF and OSF, RNF does not have a specific sensor-based failure-generation condition.

This chapter explains the dataset-defined RNF mechanism, its statistical characteristics, diagnostic limitations, possible engineering interpretations, general inspection procedures and maintenance recommendations.

6.2 Definition of Random Failure

Random Failure refers to a simulated failure event generated independently of the available operating parameters.

The AI4I dataset includes RNF to represent an unpredictable failure mechanism that cannot be directly explained using the other sensor measurements.

The failure-generation mechanism does not depend on:

- Air temperature.
- Process temperature.
- Rotational speed.
- Torque.

- Tool wear.
- Product type.

Instead, the dataset uses a random probability-based process to generate RNF events.

Important: The dataset's RNF label does not identify a specific physical component failure or provide a corresponding root cause.

6.3 Dataset-Defined RNF Generation Mechanism

According to the AI4I 2020 dataset documentation, RNF is generated using a random failure probability of 0.1%.

This means that the simulation assigns a probability of:

$$
P(RNF=1)=0.001
$$

Or:

$$
P(RNF=1)=0.1\%
$$

The probability of no random failure event is:

$$
P(RNF=0)=0.999
$$

Or:

$$
P(RNF=0)=99.9\%
$$

6.3.1 Random Failure Generation

The RNF mechanism can be conceptually represented as:

$$
RNF =
\begin{cases}
1, & \text{if random event occurs}\\
0, & \text{otherwise}
\end{cases}
$$

The random event is generated independently of the available machine operating measurements.

6.3.2 Example

Consider a simulated machine record with the following measurements:

Parameter	Value
Air temperature	300 K
Process temperature	310 K
Rotational speed	1500 rpm
Torque	40 Nm
Tool wear	100 minutes
Product type	M

These measurements do not directly determine whether RNF occurs.

The random failure-generation process independently determines the RNF label.

6.3.3 Important Interpretation

Two records with identical or very similar operating measurements may have different RNF labels.

This is because RNF is not generated using the same deterministic sensor-based conditions as the other four failure modes.

Consequently, an ML model cannot reliably infer RNF from the available operating measurements.

6.4 Statistical Characteristics of RNF

RNF is a rare failure class in the AI4I dataset.

The random generation probability is 0.1%, making positive RNF examples uncommon.

6.4.1 Expected Frequency

For a dataset containing 10,000 independent observations with a 0.1% random event probability, the expected number of RNF events is:

[
E(RNF)=N \times P(RNF=1)
]

[
E(RNF)=10000 \times 0.001
]

[
E(RNF)=10
]

This is an expected count under the random generation probability, not a guaranteed count.

The actual number of positive labels depends on the particular generated dataset.

6.4.2 Class Imbalance

Because RNF is rare, its target distribution is highly imbalanced.

For example, a hypothetical dataset with 10 RNF-positive records and 9,990 negative records would have:

Class	Number of records	Percentage
RNF = 0	9,990	99.9%
RNF = 1	10	0.1%

The actual uploaded AI4I CSV previously inspected in this project contains 19 RNF-positive records.

Therefore, the project should use the actual CSV distribution when reporting experimental results rather than assuming that every dataset version has exactly 10 positive records.

6.4.3 Impact on Machine Learning

The severe class imbalance makes RNF difficult to predict.

A model that predicts RNF = 0 for every record may achieve high accuracy because almost every record belongs to the negative class.

However, such a model would fail to detect random failure events.

Therefore, accuracy alone is not an appropriate evaluation metric for RNF.

Recommended evaluation metrics include:

- Precision.
- Recall.
- F1-score.

- Precision-Recall AUC.
- Confusion matrix.
- Balanced accuracy.

6.5 Relevant Sensor Parameters

The AI4I dataset contains several sensor measurements, but none of them directly determines RNF.

6.5.1 Air Temperature [K]

Dataset column: Air temperature [K]

Unit: Kelvin (K)

Air temperature represents the surrounding environmental temperature.

Role in RNF:

Air temperature is not directly used in the AI4I RNF generation mechanism.

It may provide general operating context but cannot reliably determine whether a random failure event occurs.

6.5.2 Process Temperature [K]

Dataset column: Process temperature [K]

Unit: Kelvin (K)

Process temperature represents the operating temperature associated with the simulated manufacturing process.

Role in RNF:

Process temperature is not directly used in the AI4I RNF generation mechanism.

6.5.3 Rotational Speed [rpm]

Dataset column: Rotational speed [rpm]

Unit: Revolutions per minute (rpm)

Rotational speed represents the operating speed of the rotating system.

Role in RNF:

Rotational speed does not determine the RNF label in the dataset.

6.5.4 Torque [Nm]

Dataset column: Torque [Nm]

Unit: Newton-metre (Nm)

Torque represents the rotational force applied to the machine's rotating system.

Role in RNF:

Torque is not directly involved in the AI4I RNF generation mechanism.

6.5.5 Tool Wear [min]

Dataset column: Tool wear [min]

Unit: Minutes

Tool wear represents accumulated cutting-tool usage.

Role in RNF:

Tool wear is not directly involved in the AI4I RNF generation mechanism.

6.5.6 Product Type

Dataset column: Type

Possible values:

- L: Low-quality variant.
- M: Medium-quality variant.
- H: High-quality variant.

Role in RNF:

Product type does not determine whether an RNF event occurs.

6.6 Engineering Interpretation of Random Failure

Random failure in real industrial environments is a broad concept.

Unexpected failures may arise from events that are not adequately represented by the currently monitored operating measurements.

Examples of possible real-world unexpected events include:

6.6.1 Sudden Component Damage

Mechanical components may experience sudden damage that is not preceded by an identifiable trend in the available sensors.

6.6.2 Electrical Interruptions

Unexpected electrical supply disturbances may interrupt machine operation.

6.6.3 Mechanical Damage

Unexpected mechanical damage may affect machine components and cause operational interruption.

6.6.4 External Disturbances

Environmental disturbances or unexpected operating events may affect equipment reliability.

Important: These are general industrial examples. They are not identified as the actual causes of the AI4I RNF labels.

The AI4I dataset does not provide the underlying physical cause of each random failure event.

6.7 Possible Engineering Symptoms

Because RNF is not associated with a specific failure mechanism in the dataset, there is no universal symptom list for its labels.

Depending on the actual equipment and failure event, possible observations may include:

Possible observation	Engineering interpretation
Unexpected machine shutdown	Possible sudden interruption
Abnormal operating noise	Possible mechanical problem
Unexpected electrical interruption	Possible supply or electrical-system issue
Sudden machine performance change	Possible unexpected operating disturbance
Component damage	Possible physical failure requiring inspection
Unexplained alarm	Requires further diagnostic investigation

These observations are general engineering examples and are not measurements or labels supplied by AI4I.

6.8 Possible Contributing Factors

Random failure events in actual industrial equipment can have different underlying causes.

Depending on the equipment, possible contributing factors may include:

6.8.1 Unexpected Component Defects

A component may fail unexpectedly due to defects or damage.

6.8.2 Electrical System Disturbances

Electrical supply interruptions or component faults may affect machine operation.

6.8.3 Mechanical Damage

Unexpected damage to mechanical components may cause sudden failure.

6.8.4 External Environmental Events

External disturbances may affect equipment operation.

6.8.5 Unobserved Operating Conditions

A failure may be associated with a condition that is not measured by the available sensors.

Dataset limitation: The AI4I RNF label is randomly generated and does not provide evidence that any particular factor caused an individual record.

6.9 Inspection Procedures

Since RNF does not identify a specific physical failure mechanism, the inspection process should begin with general equipment assessment.

The following procedures are proposed as general guidance.

Actual inspection must follow equipment-specific safety procedures and manufacturer instructions.

6.9.1 General Machine Inspection

Inspect the machine for:

- Visible physical damage.
- Unusual noise.
- Unexpected movement.
- Loose components.
- Abnormal operating behaviour.

6.9.2 Electrical System Inspection

Where applicable, qualified personnel should inspect electrical components and supply conditions using approved procedures.

6.9.3 Mechanical System Inspection

Inspect relevant mechanical components for damage, unusual wear or abnormal movement.

6.9.4 Operating History Review

Review recent machine operating records and maintenance history.

Identify whether an unexpected event occurred before the failure warning.

6.9.5 Alarm and Event Log Review

Review machine alarms, shutdown records and available diagnostic information.

These records may help identify the actual reason for an unexpected interruption.

6.10 Troubleshooting Procedures

The following table provides proposed general troubleshooting guidance for unexpected machine failures.

Observation	Possible explanation	Suggested action
Unexpected machine shutdown	Possible electrical or mechanical interruption	Review machine alarms and operating history
Unusual noise	Possible mechanical abnormality	Inspect relevant mechanical components
Sudden performance change	Possible unexpected operating disturbance	Review operating measurements and event logs
Electrical interruption	Possible electrical-system problem	Request inspection by qualified personnel
Unexplained failure warning	Insufficient diagnostic evidence	Perform a broader equipment inspection
Repeated unexpected failures	Possible recurring unidentified issue	Maintain detailed failure and inspection records

6.10.1 General Troubleshooting Workflow

Step 1: Identify the warning

Review the RNF prediction generated by the predictive maintenance system.

Step 2: Review available sensor measurements

Examine air temperature, process temperature, rotational speed, torque and tool wear.

Step 3: Review machine event logs

Check alarms, shutdown records and recent operating history.

Step 4: Perform general inspection

Inspect relevant mechanical and electrical systems according to equipment procedures.

Step 5: Identify the actual failure mechanism

Determine the physical cause using appropriate diagnostic procedures.

Step 6: Determine corrective action

Take action based on the inspection results and manufacturer guidance.

Step 7: Record the failure event

Document the observed symptoms, identified cause, corrective action and maintenance activity.

Step 8: Resume operation safely

Resume operation only after the equipment has been assessed as safe.

6.11 Preventive Maintenance Recommendations

Although random failure cannot be reliably predicted from the AI4I sensor measurements, general maintenance practices can help reduce equipment risk.

6.11.1 Periodic Equipment Inspection

Conduct periodic inspections of relevant mechanical and electrical components.

6.11.2 Alarm Monitoring

Monitor machine alarms and shutdown events to identify unexpected operating disturbances.

6.11.3 Maintenance History Management

Maintain detailed records of failures, inspections and corrective maintenance.

6.11.4 Component Condition Monitoring

Use appropriate condition-monitoring methods for equipment components that are not represented by the AI4I sensor variables.

Depending on the machine, additional monitoring may include vibration, acoustic, electrical or other condition measurements.

6.11.5 Electrical and Mechanical Safety Checks

Follow manufacturer-recommended inspection and maintenance procedures.

6.11.6 Root-Cause Investigation

Investigate unexpected failures to determine whether a recurring physical failure mechanism can be identified.

6.11.7 Sensor Coverage Improvement

Where real industrial deployment is intended, consider additional sensors to capture failure mechanisms not represented in the existing dataset.

6.11.8 Maintenance Record Management

Maintain records containing:

- Date and time of failure.
- Observed machine condition.
- Available sensor readings.
- Alarm information.
- Identified physical cause, if known.
- Inspection findings.
- Corrective maintenance actions.
- Equipment downtime.

6.12 Relationship Between RNF and Other Failure Modes

RNF differs from the other four AI4I failure mechanisms.

Failure mode	Generation mechanism
TWF	Randomly selected tool-wear threshold and simulated replacement/failure process
HDF	Temperature difference and rotational-speed conditions
PWF	Calculated mechanical power outside specified limits
OSF	Torque-wear product exceeding a product-specific threshold
RNF	Independent random event

6.12.1 Why RNF Is Different

TWF, HDF, PWF and OSF are generated using relationships involving the available operating variables.

RNF is generated independently of those variables.

Therefore, the first four failure modes contain sensor-related patterns that machine learning models may learn.

RNF does not contain the same predictable sensor relationship.

This makes RNF an important example of the limitations of sensor-based predictive maintenance.

6.13 Role of RNF in the Machine Learning System

RNF is treated as an individual binary classification target.

The model receives the available input features:

- Type.
- Air temperature.
- Process temperature.
- Rotational speed.
- Torque.
- Tool wear.

The model generates:

- RNF prediction.
- RNF probability estimate.

However, because RNF is independently generated, the model cannot reliably identify random events from these features.

6.13.1 Illustrative Prediction

Output	Illustrative value
Predicted failure type	RNF
Predicted probability	12%
Air temperature	300 K
Process temperature	310 K
Rotational speed	1500 rpm
Torque	40 Nm

Tool wear	100 minutes

These values are illustrative and do not represent an actual model prediction.

A low RNF probability does not guarantee that a random failure will not occur.

Similarly, a high probability does not establish the physical cause of an unexpected failure.

6.13.2 Model Evaluation Considerations

Because RNF is extremely rare, special care is required during model evaluation.

Recommended practices include:

- Stratified train, validation and test splitting where feasible.
- Reporting the number of positive examples in each split.
- Using precision, recall and F1-score.
- Reporting precision-recall AUC.
- Avoiding accuracy as the only evaluation metric.
- Comparing against a majority-class baseline.
- Reporting uncertainty caused by the small positive sample count.

6.13.3 Important Limitation of SMOTE and CTGAN

Synthetic oversampling can increase the number of RNF-positive training records, but it cannot create a genuine predictive relationship between RNF and the sensor measurements if none exists.

Therefore, oversampling should not be interpreted as solving the fundamental unpredictability of RNF.

For this target, it is important to distinguish class imbalance from lack of predictive information.

6.14 Role of SHAP in RNF Diagnosis

SHAP (SHapley Additive exPlanations) explains how input features influence the prediction generated by the trained machine learning model.

For RNF, SHAP may produce contributions for:

- Air temperature.
- Process temperature.
- Rotational speed.

- Torque.
- Tool wear.
- Product type.

However, because the dataset generates RNF independently of these variables, SHAP explanations should be interpreted carefully.

A model may produce feature contributions due to sampling variation, correlations, class imbalance or model behaviour.

These contributions do not establish that a particular sensor caused the random failure.

Important: SHAP can explain what influenced the model's prediction, but it cannot discover a genuine physical cause that is absent from the available information.

6.15 Role of RAG in RNF Diagnosis

The maintenance knowledge base contains general troubleshooting and inspection information that can support investigation of unexpected machine failures.

When the ML model predicts RNF, the diagnosis agent should avoid inventing a specific failure cause.

Instead, it can retrieve relevant general diagnostic procedures.

Examples of useful retrieval topics include:

- Random failure definition.
- RNF generation mechanism.
- General machine inspection.
- Electrical system inspection.
- Mechanical system inspection.
- Alarm and event log review.
- Root-cause investigation.
- Preventive maintenance procedures.

6.15.1 Example Internal Retrieval Query

Failure category: Random Failure (RNF)

Relevant information:

- Random failure characteristics.

- General machine inspection.
- Electrical and mechanical troubleshooting.
- Machine alarm history.
- Root-cause investigation.

Retrieval request:

"Retrieve general industrial machine troubleshooting procedures, electrical and mechanical inspection guidelines, unexpected shutdown investigation methods and maintenance record management practices. Do not assume a specific physical failure cause."

6.15.2 Example Retrieved Context

The RAG system may retrieve information explaining:

- The AI4I RNF generation mechanism.
- The limitations of sensor-based prediction.
- General mechanical and electrical inspection procedures.
- Methods for reviewing machine alarms and event logs.
- Recommended failure investigation practices.

The retrieved information is provided to the LLM together with the ML prediction and SHAP explanation.

The LLM should communicate that the failure mechanism is not identifiable from the available sensor measurements alone.

6.16 Faithfulness Verification Requirements

The generated diagnosis must be supported by the retrieved technical evidence.

RNF requires particular attention because an LLM may be tempted to invent a specific cause for an unexpected failure.

The faithfulness verifier should distinguish the following information:

Claim type	Required supporting evidence
RNF probability	AI4I dataset documentation
RNF independence from sensor variables	AI4I dataset documentation

General failure explanation	Engineering reference
Inspection procedure	Technical maintenance documentation
Troubleshooting recommendation	Equipment maintenance documentation
ML prediction	Actual ML model output
Feature contribution	Actual SHAP output
Actual physical root cause	Inspection findings or verified equipment records

The system must not generate unsupported claims such as:

- The motor is definitely damaged because RNF is predicted.
- High torque caused RNF.
- Increased temperature caused RNF.
- SHAP identified the actual physical cause of RNF.
- The machine will not experience random failure because the model predicted a low probability.

The system should explicitly communicate uncertainty when the available evidence is insufficient to identify the actual failure cause.

6.17 Knowledge Base Retrieval Metadata

The following metadata structure is recommended for storing RNF-related document chunks.

{

"document_id": "AI4I_MANUAL_006",

"chapter": 6,

"failure_type": "RNF",

"failure_name": "Random Failure",

"section": "Failure Generation Mechanism",

"equipment": "Simulated Industrial Milling Machine",

"source_type": "Dataset_Documentation",

"content": "Random Failure is generated independently "

"of the available operating measurements "

"using a probability-based mechanism."

}

Suggested section categories:

Category	Description
Failure_Definition	RNF explanation
Dataset_Rule	Random generation probability
Statistical_Characteristics	Class distribution and imbalance
Sensor_Parameters	Available measurements and their limitations
Engineering_Concepts	General unexpected failure mechanisms
Engineering_Symptoms	Possible failure observations
Inspection	General machine inspection
Troubleshooting	Corrective investigation
Preventive_Maintenance	General maintenance recommendations
Limitations	Prediction and dataset limitations

These categories can be used as metadata filters during vector retrieval.

6.18 Chapter Summary

Random Failure is a simulated failure mechanism in the AI4I 2020 Predictive Maintenance Dataset.

It is generated independently of the available operating measurements using a random probability of 0.1%.

Unlike TWF, HDF, PWF and OSF, RNF does not have a deterministic sensor-based failure-generation condition.

Consequently, it is difficult to predict reliably using the available AI4I features.

The proposed maintenance procedures focus on general equipment inspection, electrical and mechanical system assessment, alarm monitoring, root-cause investigation and maintenance record management.

Within the proposed research architecture:

1. The ML model estimates the likelihood of RNF, subject to its fundamental prediction limitations.
2. SHAP explains the model's feature contributions without claiming physical causality.
3. RAG retrieves relevant general inspection and troubleshooting knowledge.
4. The LLM communicates the prediction and uncertainty.
5. The faithfulness verifier checks generated claims against retrieved evidence.

This integration demonstrates an important research limitation: an explainable AI system should also recognize when the available data cannot support a reliable diagnosis.

6.19 References

1. UCI Machine Learning Repository. (2020). *AI4I 2020 Predictive Maintenance Dataset*. https://doi.org/10.24432/C5HS5C
2. Matzka, S. (2020). *Explainable Artificial Intelligence for Predictive Maintenance Applications*. Third International Conference on Artificial Intelligence for Industries (AI4I).
3. ISO 13381-1:2015. *Condition monitoring and diagnostics of machines — Prognostics — Part 1: General guidelines*.
4. ISO 17359:2018. *Condition monitoring and diagnostics of machines — General guidelines*.

Source distinction: The 0.1% random generation probability and independence from the available process measurements are based on AI4I dataset documentation. The possible physical failure examples, inspection procedures and maintenance recommendations are general engineering knowledge and proposed guidance. They are not identified causes of individual AI4I RNF records.

Chapter 7: Overall Machine Failure (Machine Failure)

Document: AI4I 2020 Predictive Maintenance Knowledge Manual
Chapter ID: AI4I_MANUAL_007
Failure Category: Overall Machine Failure
Target Variable: Machine failure
Research Project: Faithfulness-Verified Multi-Source Agentic RAG for Industrial Predictive Maintenance

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7.1 Introduction

Overall Machine Failure represents the general failure status of a machine in the AI4I 2020 Predictive Maintenance dataset. Unlike the individual failure categories discussed in previous chapters, this target does not describe one specific physical failure mechanism.

Instead, it indicates whether the machine has experienced at least one of the five defined failure modes:

1. Tool Wear Failure (TWF)
2. Heat Dissipation Failure (HDF)
3. Power Failure (PWF)
4. Overstrain Failure (OSF)
5. Random Failure (RNF)

The Machine failure column is the primary target for predicting whether a machine is operating normally or has experienced a failure.

In the proposed predictive maintenance system, this target is used by the machine learning component to estimate the overall failure risk. Individual failure-mode predictions can then provide more specific information about the possible category of failure.

7.2 Definition of Overall Machine Failure

Overall Machine Failure is a binary classification target.

It has two possible values:

Value	Meaning
0	No machine failure recorded

1 Machine failure recorded

Conceptually, the overall failure indicator can be represented as:

[
MF = TWF \lor HDF \lor PWF \lor OSF \lor RNF
]

Where:

- (MF) = Overall Machine Failure
- (TWF) = Tool Wear Failure
- (HDF) = Heat Dissipation Failure
- (PWF) = Power Failure
- (OSF) = Overstrain Failure
- (RNF) = Random Failure
- (\lor) = Logical OR operation

Therefore, if at least one failure-mode indicator is active, the overall machine failure indicator is active.

Important: Machine failure is an aggregate target, not a sixth independent failure mechanism. It indicates the presence of a failure but does not independently identify its physical cause.

7.3 Dataset Statistics

In the AI4I 2020 CSV file used in this research project, the overall machine failure distribution is:

Category	Number of records	Percentage
Normal operation (0)	9,661	96.61%
Machine failure (1)	339	3.39%
Total	10,000	100%

The dataset contains significantly more normal operating records than failure records.

This imbalance is important when developing machine learning models. A classifier that predicts every record as normal would achieve 96.61% accuracy on this particular dataset while detecting no failures.

Consequently, accuracy alone is insufficient for evaluating the predictive maintenance model.

7.4 Relationship Between Overall Failure and Individual Failure Modes

The overall failure target combines the five individual failure categories.

Failure mode	Description	Chapter reference
TWF	Failure associated with tool wear	Chapter 2
HDF	Failure associated with insufficient heat dissipation under the dataset rule	Chapter 3
PWF	Failure associated with calculated mechanical power outside the dataset range	Chapter 4
OSF	Failure associated with the torque and tool-wear interaction	Chapter 5
RNF	Randomly generated failure indicator	Chapter 6

Each individual failure mode has its own dataset-specific generation rule or mechanism.

The overall target does not contain additional information about the exact failure mechanism. It simply provides an aggregate indication that a failure has occurred.

7.4.1 Multiple Failure Modes

A single record may contain more than one active failure-mode indicator.

For example:

TWF	HDF	PWF	OSF	RNF	Machine failure
0	0	0	0	0	0
1	0	0	0	0	1
0	1	0	0	0	1
0	0	1	1	0	1
1	0	0	0	1	1

The fourth example contains two active failure-mode labels. The overall target remains 1.

In the user's dataset, 24 records contain more than one active failure-mode label.

This means that the five individual failure modes should not be treated as mutually exclusive classes. A multi-label prediction approach is more appropriate when predicting several failure indicators simultaneously.

7.5 Sensor Parameters Used for Overall Failure Prediction

The AI4I dataset contains five primary operating measurements that can be used by the predictive model.

7.5.1 Air Temperature

Air temperature represents the surrounding environmental temperature of the simulated machine.

It is measured in Kelvin (K).

It is relevant to the thermal operating environment and contributes to the temperature difference used by the HDF generation rule.

7.5.2 Process Temperature

Process temperature represents the temperature associated with the machine's operating process.

It is measured in Kelvin (K).

Together with air temperature, it helps calculate the temperature difference:

[
\Delta T = T_{process} - T_{air}
]

This derived measurement is relevant to the dataset's HDF rule.

7.5.3 Rotational Speed

Rotational speed represents the machine's rotational operating rate.

It is measured in revolutions per minute (RPM).

It contributes to mechanical power calculation and is also part of the HDF failure-generation condition.

7.5.4 Torque

Torque represents the rotational force applied by the machine.

It is measured in Newton-metres (Nm).

It contributes to mechanical power calculation and the torque-wear interaction used in the PWF and OSF dataset rules.

7.5.5 Tool Wear

Tool wear represents the accumulated tool usage in minutes.

It is measured in minutes.

It is directly relevant to the TWF generation mechanism and contributes to the OSF torque-wear interaction.

7.6 Feature Engineering for Overall Failure Prediction

Feature engineering can improve the representation of machine operating conditions.

The proposed system derives additional features from the original sensor measurements.

7.6.1 Temperature Difference

[
\Delta T = T_{process} - T_{air}
]

This feature represents the difference between process temperature and surrounding air temperature.

It can help the model learn thermal relationships relevant to HDF.

7.6.2 Mechanical Power

Mechanical power is calculated using torque and rotational speed:

[
P = \frac{2\pi \times T \times RPM}{60}
]

Where:

- (P) = Mechanical power in watts
- (T) = Torque in Newton-metres
- (RPM) = Rotational speed in revolutions per minute

This feature represents the mechanical power relationship relevant to PWF.

7.6.3 Torque-Wear Interaction

[
TW = T \times W
]

Where:

- (T) = Torque
- (W) = Tool wear

This feature represents the interaction used in the OSF generation rule.

7.6.4 Feature Engineering Summary

Feature	Calculation	Related failure mode
Temperature difference	Process temperature − Air temperature	HDF
Mechanical power	(2\pi \times T \times RPM / 60)	PWF
Torque-wear interaction	Torque × Tool wear	OSF

These engineered features are useful representations of relationships already present in the original measurements. They should not be interpreted as independent physical sensor readings.

7.7 Machine Learning-Based Overall Failure Prediction

The machine learning component predicts whether a machine is operating normally or has experienced a failure.

The prediction process consists of the following stages:

1. Receive a machine operating record.
2. Extract the sensor measurements and product type.
3. Apply the same preprocessing used during model training.
4. Calculate the engineered features.
5. Pass the processed feature vector to the trained classifier.
6. Generate a failure probability.
7. Apply a selected classification threshold.
8. Return the predicted overall failure status.

The prediction can be represented as:

$$
\hat{y} = f(X)
$$

Where:

- (X) = Processed sensor feature vector
- (f) = Trained machine learning model
- (\hat{y}) = Predicted overall failure class

The model may also generate a probability:

$$
P(MF=1 \mid X)
$$

This probability represents the model's estimated likelihood of the positive class based on the learned dataset patterns. It is not automatically a calibrated real-world probability of physical machine breakdown.

7.8 Classification Threshold and Failure Risk

Most binary classifiers produce a probability or decision score before assigning a final class.

A simple classification rule is:

$$
\hat{y} =
\begin{cases}
1 & \text{if } P(MF=1 \mid X) \geq \tau \\
0 & \text{otherwise}
\end{cases}
$$

Where (\tau) is the classification threshold.

A threshold of 0.5 is a common starting point, but it should not automatically be considered optimal for predictive maintenance.

Missing an actual failure can be more costly than investigating a false alarm. Therefore, the threshold should be selected using validation data and the intended operational trade-off.

Possible risk categories for the application's user interface are:

Model output	Suggested display
Predicted class 0	No failure predicted
Predicted class 1	Failure predicted
High probability score	Elevated predicted failure risk

If the application uses labels such as Low, Medium, or High Risk, the corresponding probability boundaries must be defined and validated by the project. They should not be presented as official AI4I categories.

7.9 Handling Class Imbalance

The dataset contains 9,661 normal records and 339 failure records.

This imbalance can cause a model to favour the majority class.

For example, a model may achieve high accuracy by predicting normal operation for most inputs while failing to identify important failure cases.

The following techniques can be considered:

7.9.1 Class Weighting

Class weighting increases the penalty associated with incorrectly classifying minority-class records.

For example, a class-balanced Random Forest can assign more importance to failure records during training.

7.9.2 Threshold Tuning

The classification threshold can be adjusted using validation predictions to improve the desired balance between precision and recall.

7.9.3 Oversampling

Techniques such as SMOTE may be evaluated as experimental approaches to improve minority-class learning.

However, oversampling must be applied only to training data. Validation and test sets should retain their original distributions.

7.9.4 Appropriate Evaluation Metrics

The following metrics should be considered:

- Precision
- Recall

- F1-score
- ROC-AUC
- PR-AUC
- Balanced accuracy
- Confusion matrix

Recall is particularly important when the objective is to reduce missed failures. Precision is important when false alarms create unnecessary inspections or maintenance costs.

PR-AUC is useful for evaluating positive-class performance when failure records are relatively rare.

7.10 Evaluation of the Overall Failure Model

The model should be evaluated on data that was not used during training or model selection.

A suitable experimental structure is:

Dataset split	Purpose
Training set	Learn model parameters
Validation set	Select models and tune thresholds
Test set	Estimate final model performance

Preprocessing operations, including imputation, encoding, and scaling, must be fitted using training data only.

The final test set should not be used repeatedly to select models or tune thresholds.

A confusion matrix provides four important outcomes:

Outcome	Meaning
True Positive (TP)	A failure record correctly predicted as failure
True Negative (TN)	A normal record correctly predicted as normal
False Positive (FP)	A normal record incorrectly predicted as failure
False Negative (FN)	A failure record incorrectly predicted as normal

In predictive maintenance, false negatives deserve particular attention because they represent failures that the model did not identify.

7.11 SHAP-Based Explainability

SHAP (SHapley Additive exPlanations) can be used to explain the predictions of the trained overall failure model.

It estimates how individual input features contribute to a particular prediction relative to a model reference value.

For a given machine record, SHAP can identify which measurements contributed positively or negatively to the model's failure prediction.

Possible explanatory features include:

- Tool wear
- Torque
- Rotational speed
- Air temperature
- Process temperature
- Temperature difference
- Mechanical power
- Torque-wear interaction
- Product type

The exact features shown depend on the model's actual input representation.

7.11.1 Local Explanation

A local SHAP explanation describes one specific prediction.

For example, a prediction may be influenced by high tool wear, a particular torque value, and an operating speed.

The explanation should report the actual SHAP values generated for that record rather than inventing a reason for the prediction.

7.11.2 Global Explanation

A global SHAP summary can identify which features have the greatest influence across a collection of predictions.

It can help researchers understand the general behaviour of the trained model.

7.11.3 Important Limitation

SHAP explains model behaviour, not necessarily the actual physical cause of a machine failure.

For example, a high SHAP contribution from tool wear does not independently prove that the physical tool has failed.

Furthermore, engineered features are derived from original measurements. If both the original variables and their engineered combinations are included in the model, their SHAP contributions may overlap or be distributed across correlated features.

Therefore, explanations must be interpreted carefully and should not be presented as proof of causation.

7.12 Role of the Maintenance Knowledge Manual

The machine learning model predicts the overall failure status, but the prediction alone does not provide complete maintenance instructions.

For example, the model may identify an elevated probability of machine failure without identifying which component requires inspection.

The maintenance knowledge manual provides supporting information about the five failure categories.

The proposed system can use the following chapter mapping:

Predicted failure category	Knowledge source
TWF	Chapter 2
HDF	Chapter 3
PWF	Chapter 4
OSF	Chapter 5
RNF	Chapter 6
General or unidentified failure	Chapter 7

The knowledge manual provides:

- Failure definitions
- Dataset-specific generation rules
- Relevant sensor relationships
- General engineering interpretations
- Inspection recommendations

- Troubleshooting procedures
- Preventive maintenance guidance
- Limitations and uncertainty statements

The manual is a research knowledge resource created for this project. It is not an official manufacturer maintenance manual.

7.13 Role of Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation connects the predictive output with relevant maintenance knowledge.

When the overall failure model predicts a failure, the RAG component retrieves relevant passages from the knowledge base.

A possible retrieval workflow is:

1. Receive the overall failure prediction.
2. Obtain individual failure-mode predictions, if available.
3. Obtain the SHAP explanation summary.
4. Construct an internal retrieval query.
5. Search the vector database for relevant maintenance passages.
6. Retrieve the most relevant knowledge chunks.
7. Provide the retrieved evidence and prediction context to the LLM.
8. Generate a structured maintenance explanation.

For example, an internal query could be:

"Overall machine failure predicted. Examine the predicted failure modes, relevant sensor measurements, model explanation, and maintenance guidance for the identified failure categories."

The retrieval query can be generated automatically by the application. The user does not need to manually write a prompt for every prediction.

7.13.1 Retrieval When the Failure Mode Is Known

If the individual classifiers predict TWF and OSF, the system can retrieve relevant passages from Chapters 2 and 5.

The LLM can then combine the available evidence to produce a maintenance explanation.

7.13.2 Retrieval When the Failure Mode Is Unknown

If the overall model predicts failure but the individual failure-mode classifiers do not identify a sufficiently reliable category, the system should not invent one.

Instead, it can retrieve general failure-triage information from Chapter 7 and explain that the precise failure mode has not been established.

This distinction is important because an aggregate failure prediction does not automatically identify the underlying failure category.

7.14 Multi-Source Context Integration

The proposed architecture combines several information sources.

Source	Information provided	Purpose
AI4I sensor record	Operating measurements	Represent machine operating conditions
Overall ML model	Failure prediction and score	Estimate overall failure status
Failure-mode models	TWF, HDF, PWF, OSF and RNF predictions	Provide more specific failure indicators
SHAP	Feature contribution values	Explain model behaviour
Knowledge manual	Maintenance information	Provide supporting technical evidence
RAG retriever	Relevant document chunks	Supply contextual information
LLM	Generated explanation	Combine predictions and retrieved evidence
Faithfulness verifier	Evidence-support assessment	Check generated claims

The LLM should receive structured information rather than only a natural-language statement that the machine has failed.

A structured context could contain:

- Overall predicted class
- Overall failure probability or score
- Individual failure-mode predictions
- Relevant sensor values

- SHAP feature contributions
- Retrieved knowledge passages
- Source identifiers and metadata

This allows the LLM to generate an explanation based on the actual prediction and retrieved evidence.

7.15 Faithfulness Verification

A major component of the proposed research is verifying whether the generated maintenance explanation is supported by the supplied evidence.

An LLM may generate statements that sound technically correct but are not supported by the available sensor data or retrieved documents.

The faithfulness verifier is intended to identify these unsupported statements.

7.15.1 Verification Process

The verification process can follow these stages:

1. Receive the LLM-generated maintenance explanation.
2. Divide the explanation into individual factual claims.
3. Identify the evidence associated with each claim.
4. Compare each claim against the retrieved passages and structured prediction context.
5. Determine whether the claim is supported, partially supported, unsupported, or contradictory.
6. Flag unsupported statements.
7. Produce a verified explanation or request regeneration.

7.15.2 Example

Model output:

"Overall machine failure is predicted. The individual failure-mode model indicates a possible HDF condition."

Retrieved evidence:

The HDF chapter describes the dataset-specific relationship between temperature difference and rotational speed.

Supported explanation:

"The predicted HDF category is associated with the thermal operating conditions represented in the AI4I dataset. The relevant sensor measurements can be reviewed against the dataset's HDF generation rule."

Unsupported explanation:

"The machine's cooling fan is physically damaged and must be replaced immediately."

The second statement is unsupported because the dataset does not contain a cooling-fan condition or a physical inspection result.

The verifier should therefore prevent the system from presenting such a statement as an established diagnosis.

7.15.3 Verification Limitations

Faithfulness verification does not guarantee that the retrieved manual is physically correct for every real machine.

It checks whether generated statements are supported by the supplied evidence. The quality of the final result still depends on the quality, relevance, and accuracy of the knowledge base.

7.16 Maintenance Recommendations

When an overall failure is predicted, the system can provide a general diagnostic workflow.

The following recommendations are general engineering guidance, not direct physical observations from the AI4I dataset.

7.16.1 Review Operating Measurements

Examine:

- Air temperature
- Process temperature
- Temperature difference
- Rotational speed
- Torque
- Tool wear
- Mechanical power
- Torque-wear interaction

Check whether the measurements are plausible and whether they indicate conditions associated with any of the dataset's failure categories.

7.16.2 Identify Possible Failure Categories

Review the individual failure-mode predictions.

If one or more categories are predicted, inspect the corresponding knowledge manual chapters.

If the failure category remains uncertain, avoid assigning a definitive physical cause.

7.16.3 Perform Physical Inspection

For an actual industrial machine, qualified maintenance personnel should inspect the relevant components using the manufacturer's approved procedures.

Possible inspection areas include:

- Tool condition
- Cooling and heat-dissipation arrangements
- Drive and power transmission components
- Mechanical loading
- Sensors and measurement connections

The specific inspection procedure must be selected according to the actual machine and its manufacturer's documentation.

7.16.4 Record Maintenance Findings

Maintenance personnel should record:

- Inspection date
- Observed machine condition
- Identified fault, if any
- Corrective action
- Replaced components
- Post-maintenance measurements
- Final operating status

These records can support future analysis and help compare model predictions with actual maintenance outcomes.

7.17 Preventive Maintenance Strategy

A preventive maintenance strategy aims to reduce unexpected machine downtime through regular monitoring and inspection.

For an actual industrial implementation, the following general practices may be considered:

1. Monitor machine operating measurements continuously or at suitable intervals.
2. Review abnormal temperature and mechanical operating conditions.
3. Inspect tools according to usage and manufacturer recommendations.
4. Monitor changes in torque and rotational speed.
5. Check cooling arrangements and mechanical components.
6. Investigate repeated model warnings.
7. Maintain a record of inspections and corrective actions.
8. Reassess model performance using verified maintenance outcomes.

The AI4I dataset does not define a universal maintenance schedule. Inspection intervals must be established using actual machine requirements, operating conditions, and manufacturer recommendations.

7.18 Integration With the Proposed Research Architecture

Overall Machine Failure is the central binary prediction task in the proposed system.

The complete workflow is:

Stage 1: Data Input

The user or connected data source provides machine operating measurements.

Stage 2: Preprocessing

The application validates the input, handles categorical variables, and applies the preprocessing pipeline.

Stage 3: Feature Engineering

The system calculates temperature difference, mechanical power, and torque-wear interaction.

Stage 4: Overall Failure Prediction

The trained machine learning model predicts the overall failure class and generates a probability or decision score.

Stage 5: Failure-Mode Prediction

Separate classifiers estimate the individual failure categories.

Stage 6: Explainability

SHAP generates feature-level explanations for the relevant prediction models.

Stage 7: Knowledge Retrieval

The RAG component retrieves relevant passages from the maintenance knowledge base.

Stage 8: LLM-Based Explanation

The LLM combines the structured prediction context, SHAP explanation, and retrieved maintenance evidence.

Stage 9: Faithfulness Verification

The verifier checks whether the generated claims are supported by the supplied evidence.

Stage 10: User Output

The application displays the prediction, supporting explanation, evidence references, uncertainty, and appropriate maintenance guidance.

7.19 Suggested Output Format in the Application

The application can present an overall failure result using a structured interface.

Output field	Example content
Overall status	Failure predicted
Prediction score	Actual model-generated score
Predicted failure modes	Actual outputs from individual classifiers
Important features	Actual SHAP results
Supporting knowledge	Retrieved manual passages
Explanation	LLM-generated, verified explanation
Evidence status	Supported / partially supported / unsupported
Maintenance guidance	Evidence-grounded recommendations

The example labels above describe the proposed interface. Actual scores, predicted categories, and explanations must be generated from the running system.

7.20 Metadata for RAG Knowledge Base

The chapter can be stored in the vector database as a collection of smaller semantic chunks.

Each chunk should preserve its source and chapter metadata.

Example metadata:

Metadata field	Value
document_id	AI4I_MANUAL_007
document_title	AI4I 2020 Predictive Maintenance Knowledge Manual
chapter_number	7
chapter_title	Overall Machine Failure
failure_type	Overall Machine Failure
source_type	Project-created technical knowledge manual
dataset	AI4I 2020 Predictive Maintenance
target_column	Machine failure
content_category	Overall failure prediction
version	1.0

A chunk can contain a focused topic, such as the definition of overall machine failure, class imbalance, SHAP interpretation, or RAG verification.

This metadata enables the retriever to filter and identify the relevant knowledge source.

7.21 Important Limitations and Interpretation Guidelines

The following limitations must be considered when interpreting the overall failure prediction.

7.21.1 Synthetic Dataset

AI4I 2020 is a synthetic predictive maintenance dataset. Its failure labels are generated using predefined simulation mechanisms.

The dataset does not represent direct measurements from a particular commercial machine.

7.21.2 Aggregate Target

The Machine failure column identifies the presence of a failure but does not independently identify its exact category.

Individual failure-mode predictions are required for more specific classification.

7.21.3 Overlapping Failure Categories

More than one failure-mode indicator can be active for a record.

The individual categories should therefore not be interpreted as mutually exclusive classes.

7.21.4 Class Imbalance

The failure class represents only 3.39% of the records in the user's CSV.

Evaluation should include minority-class metrics rather than relying only on accuracy.

7.21.5 Model Prediction Is Not Physical Confirmation

A model prediction is an estimate based on learned dataset patterns.

It does not replace physical inspection, maintenance expertise, or manufacturer-approved diagnostic procedures.

7.21.6 SHAP Is Not Causal Proof

SHAP describes how model features contribute to a prediction. It does not independently establish the physical cause of failure.

7.21.7 RAG Evidence Quality

The quality of generated maintenance explanations depends on the retrieved documents and their technical accuracy.

The project-created knowledge manual should not be treated as an official machine manufacturer's service document.

7.22 Chapter Summary

Overall Machine Failure is the aggregate binary target in the AI4I 2020 Predictive Maintenance dataset. It indicates whether at least one of the five defined failure modes has occurred.

In the user's dataset, 339 out of 10,000 records are labelled as failures, while 9,661 records represent normal operation.

The overall failure classifier provides an initial estimate of machine failure status. Individual failure-mode classifiers can provide additional information about possible failure categories, while SHAP helps explain the behaviour of the predictive models.

The RAG component retrieves relevant maintenance knowledge from the project-created technical manual. The LLM uses this information alongside the model outputs and SHAP explanations to generate a maintenance-oriented response.

Finally, the faithfulness verifier checks whether generated claims are supported by the available evidence.

This chapter establishes the overall machine failure prediction and explanation workflow that connects the individual failure-mode chapters with the proposed multi-source agentic RAG architecture.

References

1. AI4I 2020 Predictive Maintenance Dataset. UCI Machine Learning Repository. https://doi.org/10.24432/C5HS5C
2. Matzka, S. (2020). *Explainable Artificial Intelligence for Predictive Maintenance Applications*.
3. Waghulde et al. (2025). *Evaluating Machine Learning and Deep Learning Models for Predictive Maintenance: A Study Using the AI4I 2020 Dataset*.
4. Ibtahazzaman et al. (2025). *Enhancing Predictive Maintenance in Smart Textile Manufacturing Using Machine Learning Models*.

Reference note: The dataset is the source for the AI4I-specific target and failure-generation concepts. The project-created manual's general inspection and maintenance recommendations are supporting engineering guidance and should be validated against the documentation of the actual industrial equipment before deployment.